Top Curr Chem (Z) (2016) 374:55
DOI 10.1007/s41061-016-0053-z

REVIEW

Nickel-Catalyzed Aromatic C–H Functionalization

Junichiro Yamaguchi[1] · Kei Muto[1] · Kenichiro Itami[2,3]

Received: 13 May 2016 / Accepted: 11 July 2016 / Published online: 1 August 2016

Abstract Catalytic C–H functionalization using transition metals has received significant interest from organic chemists because it provides a new strategy to construct carbon–carbon bonds and carbon–heteroatom bonds in highly functionalized, complex molecules without pre-functionalization. Recently, inexpensive catalysts based on transition metals such as copper, iron, cobalt, and nickel have seen more use in the laboratory. This review describes recent progress in nickel-catalyzed aromatic C–H functionalization reactions classified by reaction types and reaction partners. Furthermore, some reaction mechanisms are described and cutting-edge syntheses of natural products and pharmaceuticals using nickel-catalyzed aromatic C–H functionalization are presented.

Keywords C–H functionalization · C–H activation · Nickel · Heteroarenes · Arenes · Catalyst

This article is part of the Topical Collection "Ni- and Fe-Based Cross-Coupling Reactions", edited by Arkaitz Correa.

✉ Junichiro Yamaguchi
junyamaguchi@waseda.jp

✉ Kenichiro Itami
itami@chem.nagoya-u.ac.jp

1 Department of Applied Chemistry, Waseda University, Tokyo 169-8555, Japan

2 Institute of Transformative Bio-Molecules (WPI-ITbM) and Graduate School of Science, Nagoya University, Chikusa, Nagoya 464-8602, Japan

3 JST-ERATO, Itami Molecular Nanocarbon Project, Nagoya University, Chikusa, Nagoya 464-8602, Japan

1 Introduction

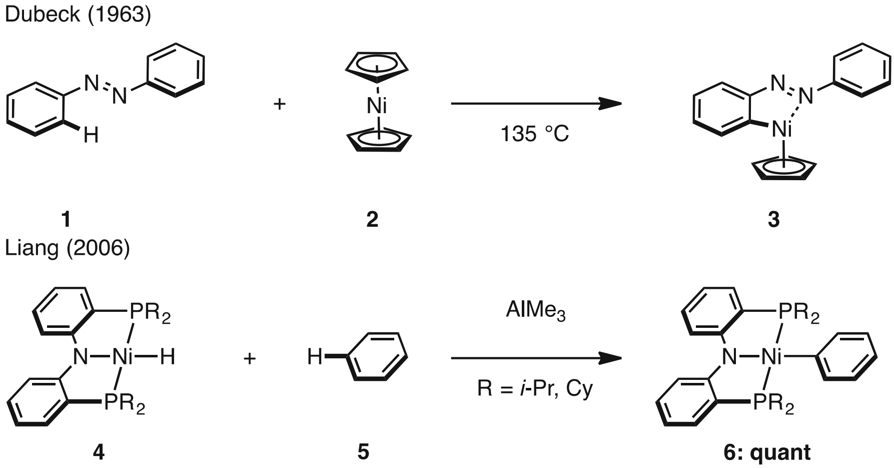
C–H functionalization using nickel catalysts has recently received significant attention from the field of organic chemistry because nickel catalysts are less expensive than noble transition metal catalysts and less toxic than commonly used metal catalysts such as palladium catalysts [1–7]. Important contributions to C–H activation of aromatic compounds by nickel were made as early as the 1960s. In 1963, Dubeck and coworkers prepared an oxidative adduct of a C–H bond on an aromatic ring onto a nickel complex (Scheme 1) [8]. When azobenzene (**1**) was mixed with nickelocene (**2**) at 135 °C, C–H nickelation at the *ortho* C–H bond of **1** occurred to give nickel complex **3** by using the diazo moiety as a directing group (chelation-assisted group). Thereafter, there were no reports describing the C–H nickelation of aromatic C–H bonds for more than 50 years. In 2006, Liang and coworkers reported that pincer nickel complex **4** could react with benzene (**5**) to furnish complex **6**, which likely occurred via the oxidative addition of the C–H bond of benzene without any directing group [9].

This review focuses on nickel-catalyzed C–H functionalization, particularly on aromatic rings (aromatic C–H functionalization), and showcases recent examples classified by reaction types and reaction partners. Furthermore, some reaction mechanisms are described and cutting-edge syntheses of natural products and pharmaceuticals using nickel-catalyzed aromatic C–H functionalization are presented.

2 Redox-Neutral Aromatic C–H Functionalization with Organohalides

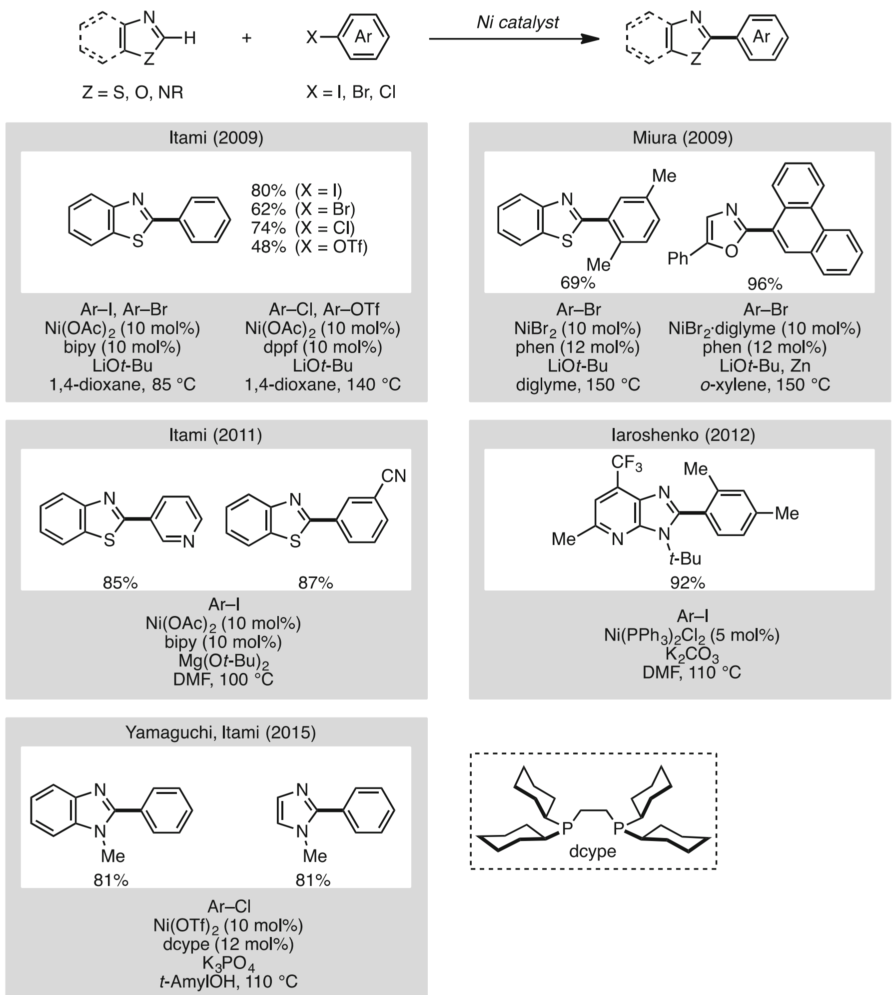
2.1 Aromatic C–H Arylation

Heterobiaryl frameworks such as 2-aryl-1,3-azoles represent privileged structural motifs often found or used in bioactive molecules. In 2009, the groups of Itami and



Scheme 1 Pioneering work for C–H activation using nickel complexes

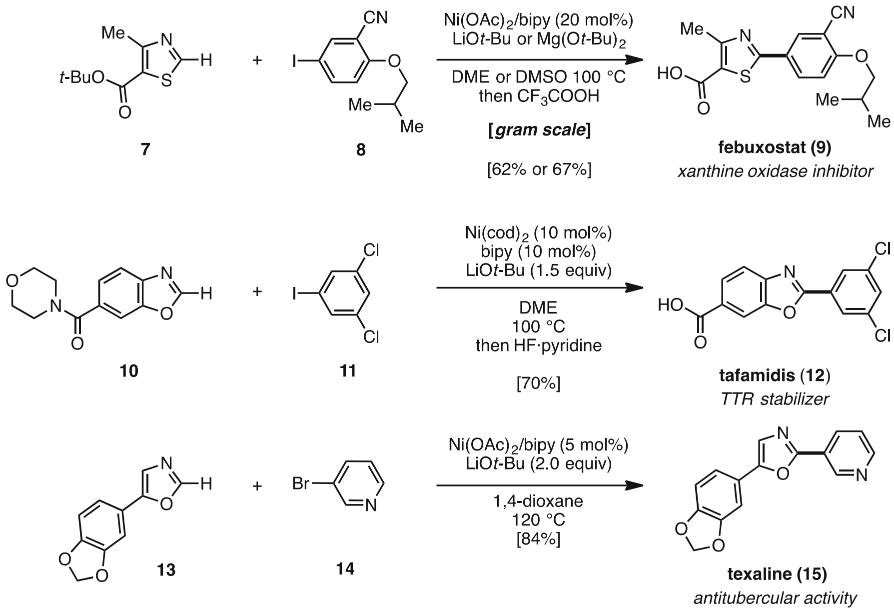
Miura independently reported the first example of nickel-catalyzed C–H arylation of 1,3-azoles with aryl halides to synthesize 2-aryl-1,3-azoles (Scheme 2) [10, 11]. Itami and coworkers developed a [$Ni(OAc)_2$/2,2′-bipyridyl (bipy)/LiO*t*-Bu] catalyst system for the C–H arylation of 1,3-azoles with aryl halides, while Miura simultaneously reported similar conditions [$NiBr_2$/1,10-phenanthroline (phen)/LiO*t*-Bu] for the C–H arylation between 1,3-(benzo)azoles and aryl bromides. In both of these reaction conditions, it was essential to use LiO*t*-Bu as the base. Thereafter, a new protocol for $Ni(OAc)_2$/bipy-catalyzed C–H arylation using $Mg(Ot\text{-}Bu)_2$ as a milder and less expensive alternative to LiO*t*-Bu was developed in 2011 by Itami and coworkers [12]. Compared with the protocol that uses LiO*t*-Bu, the newly developed $Mg(Ot\text{-}Bu)_2$-based method is more effective for reactions involving



Scheme 2 C–H arylation of 1,3-azoles with aryl halides

electron-deficient haloarenes. Then, in 2012, the Iaroshenko group reported that imidazo[4,5-*b*]pyridines can be reacted with aryl iodides or aryl bromides in the presence of $Ni(PPh_3)_2Cl_2$ as the catalyst and K_2CO_3 in DMF to give the corresponding coupling products [13]. Although these reactions can be performed with palladium catalysts, yields using the nickel catalyst were superior. In general, benzimidazoles and imidazoles had not been viable substrates in nickel-catalyzed C–H arylation. Additionally, aryl chlorides had not been commonly used for the nickel-catalyzed C–H arylation of 1,3-azoles. However, in 2015, Itami, Yamaguchi, and coworkers discovered optimal conditions for the C–H arylation of benzimidazoles and imidazoles with aryl chlorides [14]. The key to achieving the C–H arylation of imidazoles is to use *t*-AmylOH as the solvent with a Ni/1,2-bis(dicyclohexylphosphino)ethane (dcype) catalyst (for reactions involving other aryl sources under the same catalytic conditions, see Sect. 4.1).

These methods for the nickel-catalyzed C–H arylation of 1,3-azoles can be applied to the synthesis of biologically active molecules (Scheme 3) [10, 12]. Under the influence of a $Ni(OAc)_2$/bipy catalyst and LiO*t*-Bu, thiazole **7** and aryl iodide **8** underwent coupling in dimethoxyethane (DME) at 100 °C to furnish the corresponding coupling product. A $Mg(Ot\text{-}Bu)_2$/dimethylsulfoxide (DMSO) system was equally effective for the coupling. Subsequent treatment with CF_3CO_2H afforded febuxostat (**9**) [15], a xanthine oxidase inhibitor, in 62–67 % overall yield. Following the present synthetic route, only three steps are required (in 43–46 % overall yield) to accomplish the synthesis of **9**. Additionally, this reaction was applicable for the synthesis of tafamidis (**12**), a transthyretin (TTR) stabilizer [16].



Scheme 3 Syntheses of febuxostat (**9**), tafamidis (**12**), and texaline (**15**) by Ni-catalyzed arylation of azoles

Carboxamide-substituted benzoxazole **10** was coupled with 1,3-dichloro-5-iodobenzene (**11**) to furnish the corresponding product. Subsequent hydrolysis of the amide using HF·pyridine provided **12** in 70 % overall yield. Additionally, it was found that texaline (**15**) [17], an antimycobacterial alkaloid, could be obtained in 84 % yield by coupling oxazole **13** with aryl bromide **14** by the action of the $Ni(OAc)_2$/bipy/LiO*t*-Bu catalytic system in 1,4-dioxane at 120 °C. The efficient and rapid syntheses of febuxostat (**9**), tafamidis (**12**), and texaline (**15**) highlight the potential of the present nickel catalysis in a range of synthetic applications.

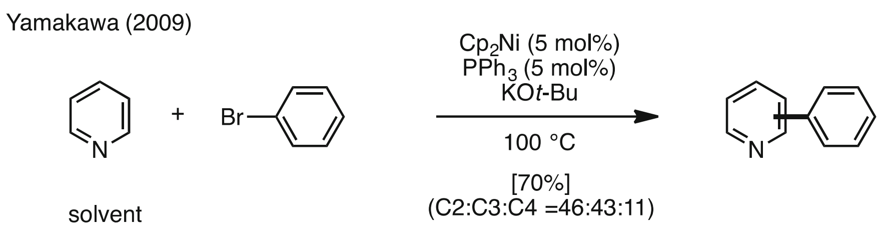
Concomitantly with the above C–H arylation of 1,3-azoles, Yamakawa and coworkers reported the direct arylation of simple benzene, naphthalene, and pyridine with aryl bromides using a Cp_2Ni/BEt_3 (or PPh_3)/KO*t*-Bu catalyst system (Scheme 4) [18]. It was presumed that the cross-coupling reaction proceeds through a radical-type pathway, because regioisomers were produced by this nickel-catalyzed system.

These nickel-catalyzed transformations work not only for heteroarenes, but also for arenes (benzene derivatives). In 2014, Chatani and coworkers reported a nickel-catalyzed C–H arylation of benzamides with aryl iodides (Scheme 5) [19]. For example, benzamide **16** bearing a quinoline moiety as a chelation-assisted group was coupled with iodobenzene (**17**) in the presence of catalytic $Ni(OTf)_2$ to afford the corresponding coupling product **18** in 92 % yield with exclusive *ortho*-selectivity.

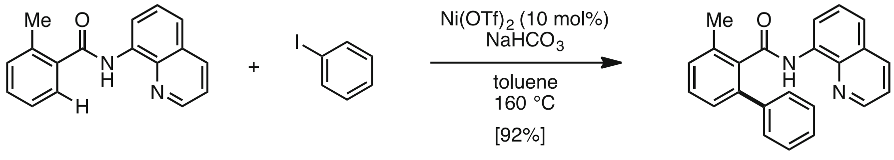
2.2 Aromatic C–H Alkynylation

The first example of C–H alkynylation of heteroarenes was reported by Miura and coworkers (Scheme 6) [20]. For example, aryloxazole **19** was reacted with bromophenylethyne (**20**) using a nickel catalyst [$Ni(cod)_2$/1,2-bis(diphenylphosphino)benzene (dppbz)] and LiO*t*-Bu as base to afford the corresponding alkynyl adduct **21** in 91 % yield.

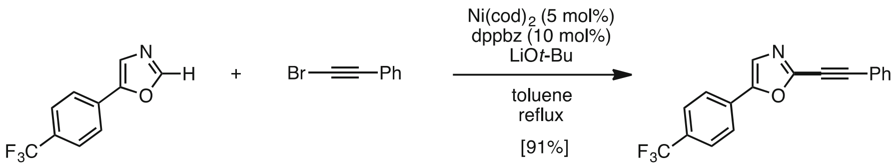
In 2015, several examples of C–H alkynylation of six-membered (hetero)arenes with alkynyl bromides were published using chelation-assisted groups (Scheme 7) [21–23]. The groups of Shi (aromatic amides containing 2-pyridyldimethylamine), Li (aromatic amides containing 8-aminoquinoline), and Balaraman (aromatic amides containing 8-aminoquinoline) independently accomplished nickel-catalyzed C–H alkynylations of pyridylamide and benzamide derivatives with bromoethynes to give the corresponding *ortho*-substituted products in excellent yields.



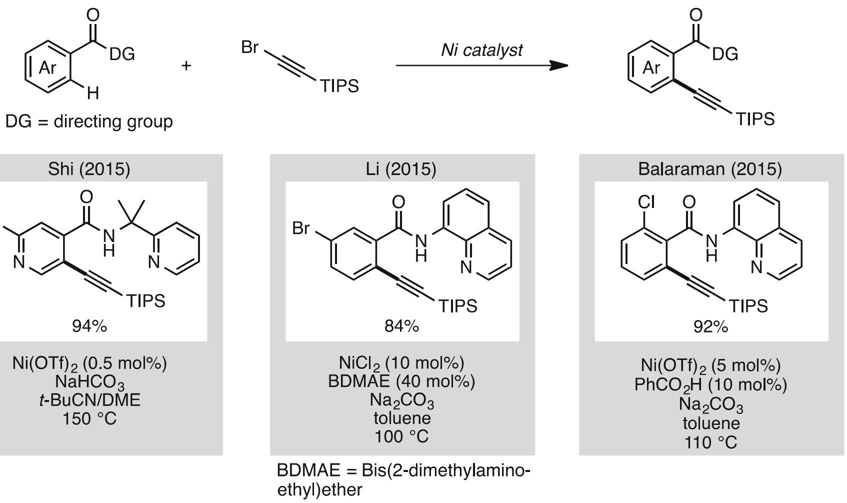
Scheme 4 C–H arylation of pyridine with aryl halides



Scheme 5 Chelation-assisted C–H arylation of benzamides

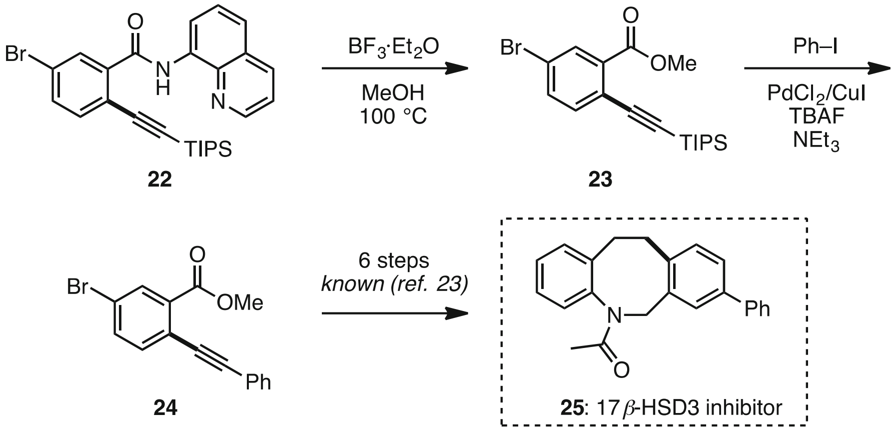


Scheme 6 C–H alkynylation of azoles with alkynyl bromides

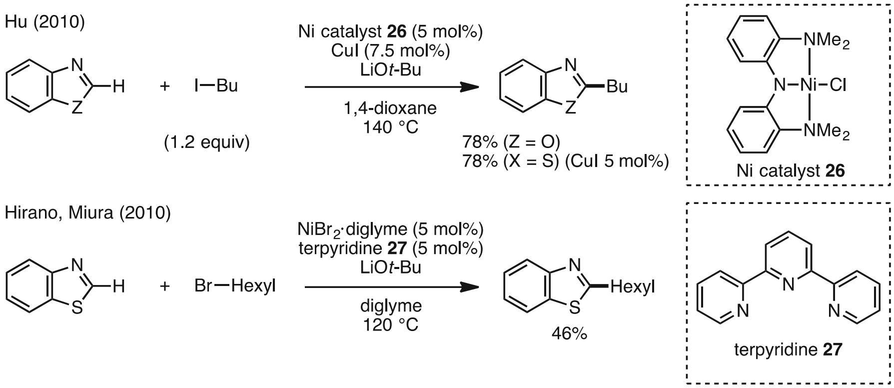


Scheme 7 Aromatic C–H alkynylations with alkynyl halides

The Li group also applied their method to synthesize an intermediate of a 17β-hydroxysteroid dehydrogenase type-3 (17β-HSD3) inhibitor, which was considered to be a clinical candidate for the treatment of prostate cancer (Scheme 8) [22]. The obtained alkynyl product **22** was treated with $BF_3 \cdot OEt_2$ to give ester **23**, followed by sila-Sonogashira coupling with iodobenzene to give diarylethyne **24**, which was an intermediate toward 17 β-HSD3 inhibitor **25** [24].



Scheme 8 Synthesis of a 17β-HSD3 inhibitor



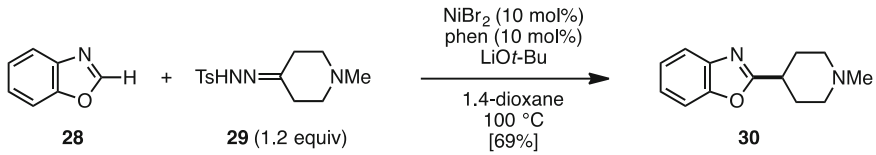
Scheme 9 C–H alkylation of 1,3-azoles and alkyl halides

2.3 Aromatic C–H Alkylation

In 2010, C–H alkylations of 1,3-azoles with alkyl halides were reported independently by the groups of Hu and Miura (Scheme 9) [25, 26]. C–H activation using alkyl reactants is relatively rare compared with using unsaturated reactants because of the difficulty in suppressing the β-hydrogen elimination of alkylmetal reagents. To suppress β-hydrogen elimination, Hu's group used a pincer-type nickel catalyst **26** in the presence of CuI as a co-catalyst. Meanwhile, the Hirano and Miura group utilized terpyridine **27** as the ligand.

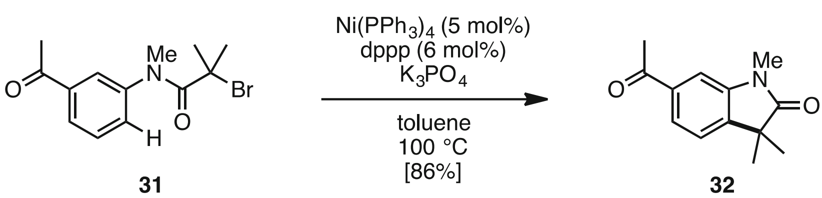
Although technically the reaction does not involve alkyl halides, the Miura group developed a C–H alkylation of 1,3-azoles with alkyl hydrazones as the alkylating agent in 2012 (Scheme 10) [27]. *N*-Tosylhydrazone **29** was converted to the diazo species in situ that reacted with benzoxazole (**28**) in the presence of a $NiBr_2$/phen/LiO*t*-Bu catalytic system to afford the secondary alkylated product **30** in moderate

Hirano, Miura (2012)



Scheme 10 Aromatic C–H alkylation using alkyl hydrazones

Lei (2013)



Scheme 11 Intramolecular aromatic C–H alkylation

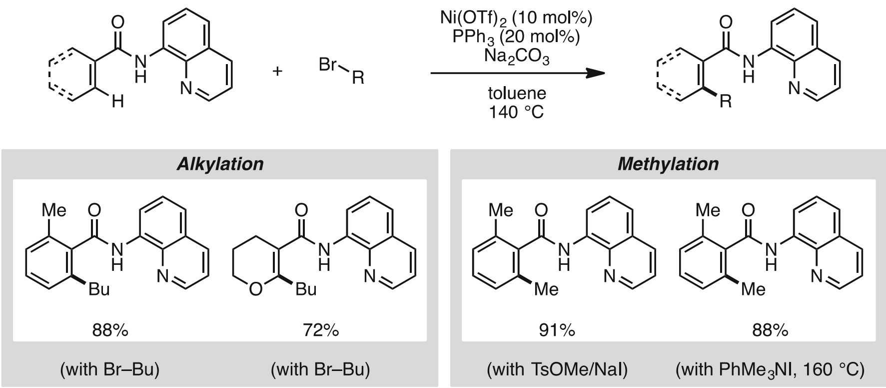
yields. It is of note that this reaction also proceeded when a cobalt catalyst was used. This C–H alkylation method shows the potential utility of alternative methods for alkylated 1,3-azoles.

Furthermore, the Lei group reported an intramolecular C–H alkylation of arenes in 2013 (Scheme 11) [28]. When α-bromoamide **31** was reacted with catalytic $Ni(PPh_3)_4$/1,3-bis(diphenylphosphino)propane (dppp) and K_3PO_4 in toluene at 100 °C, cyclization product **32** was formed in 86 % yield. It was assumed that this reaction proceeds through a radical pathway because the reaction was quenched when 2,2,6,6-tetramethylpiperidine 1-oxyl (TEMPO) was added as a radical scavenger. Additionally, this reaction worked well under visible light in the presence of a tris(2-phenylpyridinato)iridium [$Ir(ppy)_3$] photocatalyst.

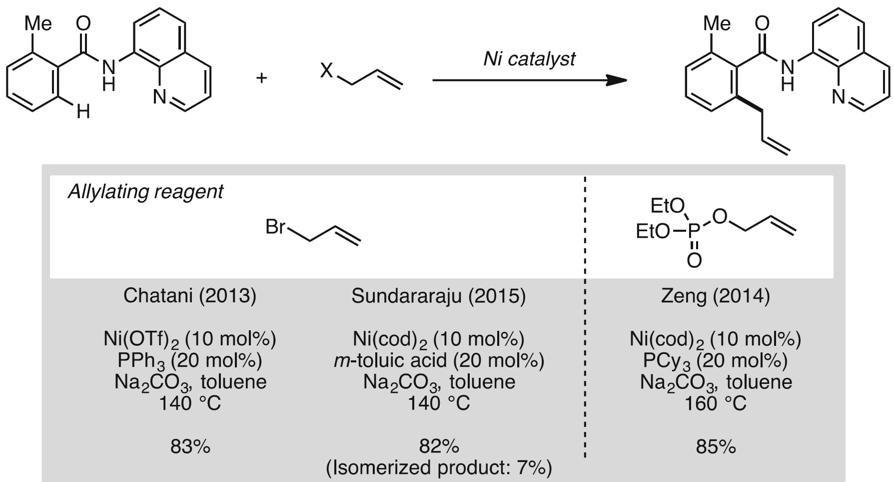
Intermolecular C–H alkylation of arenes through nickel catalysis was achieved by using a chelation-assisted group by Chatani and coworkers (Scheme 12) [29–31]. Arylamides containing 8-aminoquinoline reacted with various alkyl bromides in the presence of a $Ni(OTf)_2/PPh_3$ catalyst to give C–H alkylation products in good yields. These reaction conditions can also be applied to methylation when using TsOMe/NaI or $PhMe_3NI$ as the methylating reagent.

In 2013, Chatani's group also demonstrated a C–H allylation of arylamides using allyl bromides (Scheme 13) [29]. Thereafter, Sundararaju and coworkers reported a similar result using *m*-toluic acid as an effective additive [32]. Furthermore, in 2014, the group of Zeng obtained the same products by using allyl phosphates as allylating reagents instead of allyl bromides [33].

This chelation-assisted group, the 8-aminoquinoline moiety, enabled the use of not only primary alkyl and allyl halides but also secondary alkyl halides (Scheme 14) [34]. The group of Ackermann found that bis(2-dimethylaminoethyl)ether (BDMAE) was an effective ligand for the nickel-catalyzed C–



Scheme 12 C–H alkylation of benzamides with alkyl halides

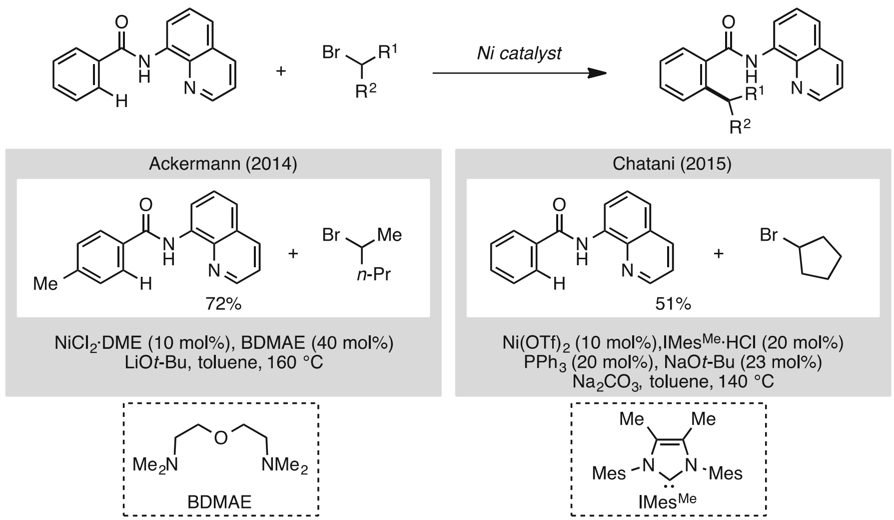


Scheme 13 C–H allylation of benzamides

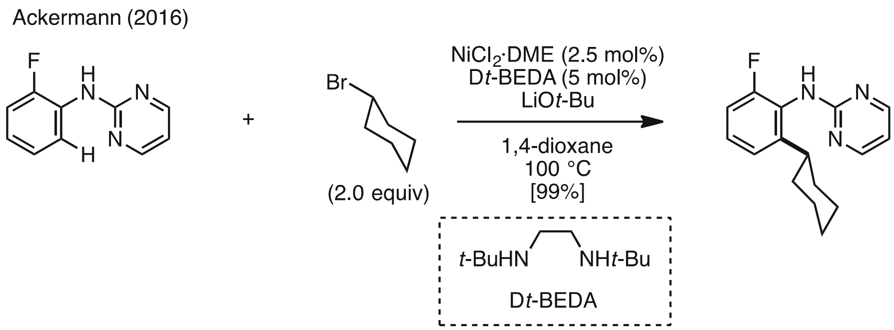
H alkylation of benzamides with secondary alkyl bromides. Although only one example was shown, Chatani and coworkers also reported a nickel-catalyzed C–H alkylation of benzamides with cyclopropyl bromide using IMes^{Me} as the ligand [30].

Very recently, Ackermann and coworkers reported a nickel-catalyzed C–H alkylation of anilines with secondary alkyl halides (Scheme 15) [35]. Use of *N*,*N*′-di-*tert*-butylethane-1,2-diamine (Dt-BEDA), a vicinal diamine, as a ligand, allowed anilines that bear a monodentate *N*-pyrimidyl substituent to be alkylated under nickel catalysis in excellent yield.

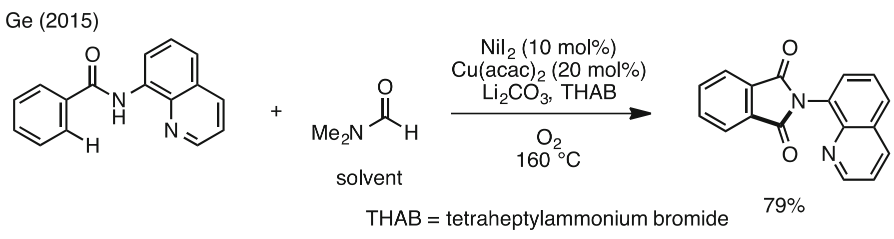
In 2014, a catalytic C–H carbonylation of arylamides was developed by Ge and coworkers (Scheme 16) [36]. Although similar types of carboxylation are known



Scheme 14 C–H alkylations of benzamides with secondary alkyl halides



Scheme 15 C–H alkylations of anilines with secondary alkyl halides



Scheme 16 C–H acylation of benzamides

with palladium, ruthenium, rhodium, and cobalt catalysts, this was the first example involving nickel/copper catalysis. Additionally, these modified conditions can be applied to aliphatic amides.

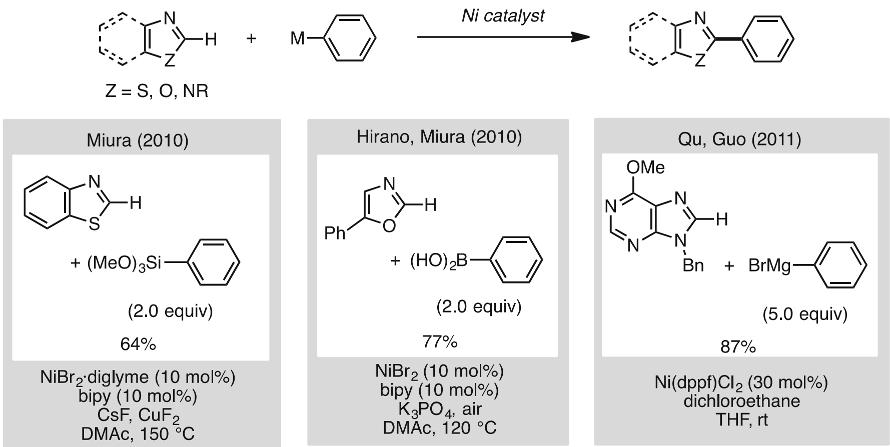
3 Oxidative Aromatic C–H Functionalization

3.1 Oxidative C–H Arylation and Alkenylation

Nickel-catalyzed oxidative C–H arylations using arylmetal species as the aryl source have also been reported in recent years (Scheme 17). The first nickel-catalyzed oxidative direct arylation was disclosed by Miura and coworkers in 2010; they developed a direct arylation of 1,3-azoles with arylsilanes in the presence of Cu salts as oxidants [37]. This report can be regarded as an important finding not only because this is the first nickel-catalyzed oxidative C–H arylation but also because oxidative C–H arylation with arylsilanes is rare even with other transition metal catalysts. A nickel-catalyzed oxidative C–H arylation of (benz)oxazoles with arylboronic acids employing air as the oxidant was also developed by the group of Miura in the same year [38]. An oxidative direct arylation using aryl Grignard reagents through a nickel catalyst system was reported by Qu, Guo, and coworkers [39].

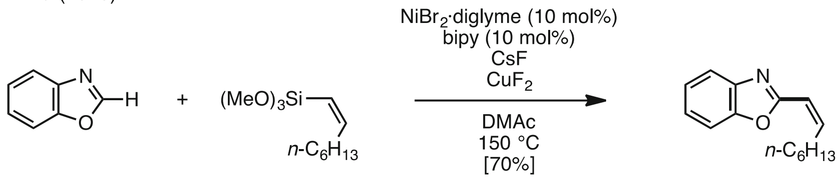
Miura's conditions can be used for the C–H alkenylation of 1,3-azoles with alkenylsilanes and alkenylboronic acids, and two examples are shown in Scheme 18 [35]. Benzoxazole (**28**) was reacted with alkenylsilane **33** under nickel catalysis to afford alkenylbenzoxazole **34** in 70 % yield. When alkenylboronic acid **36** was used as the alkenyl source, 5-phenyloxazole (**35**) could be alkenylated in the presence of a nickel catalyst to give the corresponding coupling product **37** in 47 % yield.

In 2009, Chatani, Tobisu, and coworkers reported a nickel-catalyzed C–H arylation of azines with an arylzinc species (Scheme 19) [40]. By using diarylzinc as an aryl nucleophile and $Ni(cod)_2/PCy_3$ as a catalyst, various azines such as pyridines, quinolines, phenanthridines, and pyrazines were arylated in good to excellent yields. This reaction provided C2-arylated azines regioselectively.

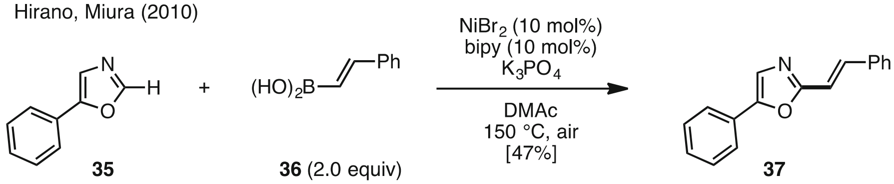


Scheme 17 Ni-catalyzed direct C–H arylation and alkenylation of azoles with organosilicon, boron, and Grignard reagents

Miura (2010)

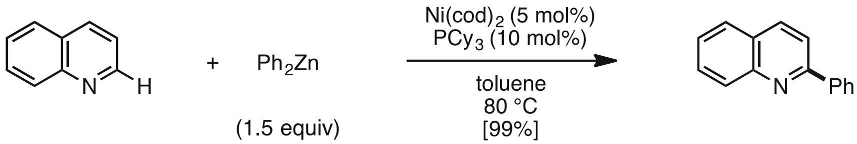


Hirano, Miura (2010)



Scheme 18 Oxidative C–H arylation of azoles with organosilicon and boron reagents

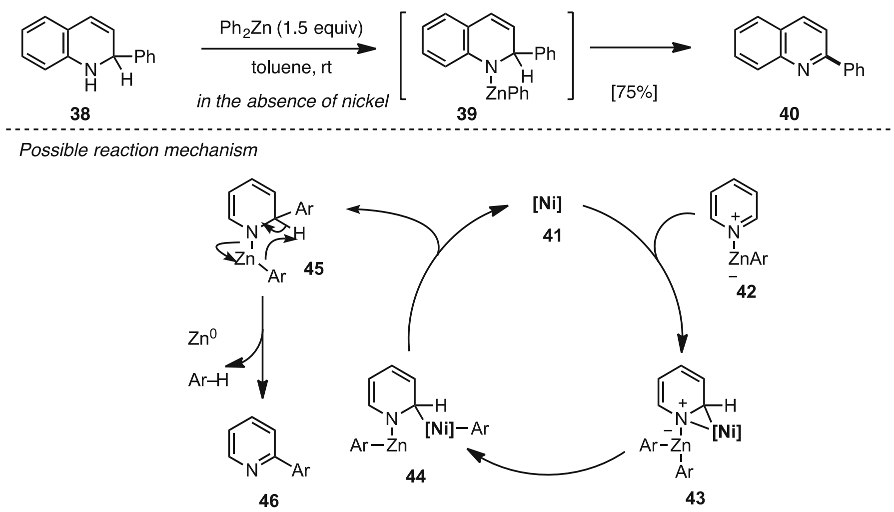
Tobisu, Chatani (2009)



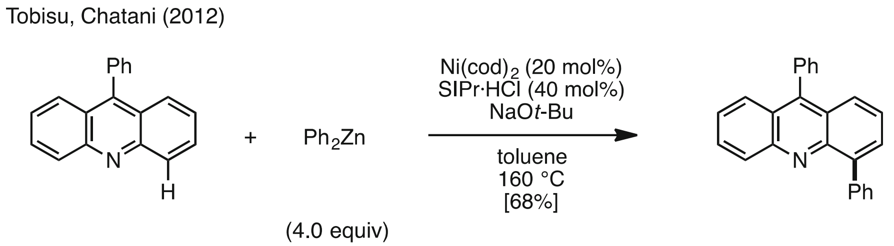
Scheme 19 C–H arylation of azines with arylzinc reagents

The mechanism of this coupling reaction is assumed to consist of addition of the aryl nucleophile under nickel catalysis, followed by in situ rearomatization to form the coupling product. To support this hypothesis, the rearomatization reaction was specifically studied by the group of Chatani [41]. Treatment of 2-phenyl-1,2-dihydroquinoline (**39**; prepared by nucleophilic addition of phenyllithium onto quinoline) with diphenylzinc at room temperature produced rearomatized product **40** without the nickel catalyst in 75 % yield (Scheme 20). According to this result, loss of the C2 hydrogen atom of **38** proceeds via the intermediacy of an organozinc species such as **39**. On the basis of this investigation, a reaction mechanism for the nickel-catalyzed C–H arylation of azines with diarylzinc was proposed. Nickel(0) species **41** initially reacts with diarylzinc–pyridine adduct **42** to form azanickelacyclopropane **43**. An intramolecular aryl transfer affords nickel species **44**, which subsequently liberates Ni^0 catalyst **41** along with zinc amide **45** by reductive elimination. The zinc amide species **45** then undergoes rapid oxidative rearomatization, leading to arylated product **46**.

Similarly, a C–H arylation of acridine at its C4 position was also reported by the Chatani group using a $Ni(cod)_2$/SIPr·HCl/NaO*t*-Bu catalytic system (Scheme 21) [42]. Under the influence of a catalytic amount of $Ni(cod)_2$, 1,3-bis(2,6-diisopropylphenyl)imidazolinium chloride (SIPr·HCl), and a stoichiometric amount



Scheme 20 Reaction mechanism for Ni-catalyzed C–H arylation of azines with arylzinc reagents



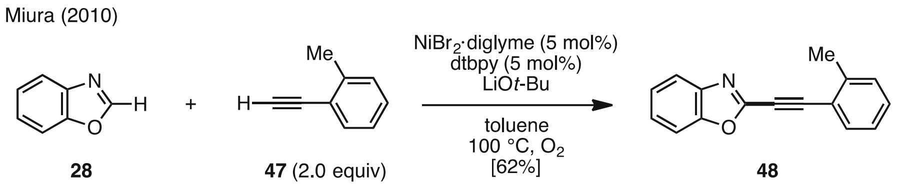
Scheme 21 C4-selective C–H arylation of acridine with arylzinc reagents

of NaO*t*-Bu, acridines were cross-coupled with excess amounts of diarylzinc reagents to afford C4-arylated products regioselectively.

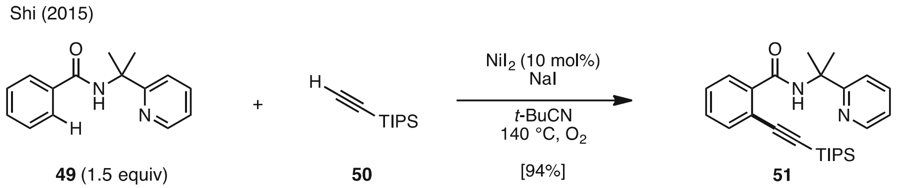
3.2 Oxidative C–H Alkynylation

Nickel-catalyzed oxidative C–H alkynylation, which involves the use of alkynes directly without pre-functionalization, was also reported by the groups of Miura and Shi. For example, in the presence of catalytic $NiBr_2$/di-*t*-butylbipyridyl (dtbpy) and LiO*t*-Bu as a base in toluene under oxygen atmosphere, benzoxazole (**28**) was reacted with arylethyne **47** to afford coupling product **48** in 62 % yield (Scheme 22) [43].

In contrast, Shi and coworkers reported in 2015 that arylamides (**49**: containing a 2-pyridyldimethylamine moiety) can react with triisopropylsilyl acetylene **50** in the presence of catalytic NiI_2 and NaI as an additive in *t*-BuCN (similar reaction conditions to Scheme 7) under an oxygen atmosphere to give C–H alkynylated product **51** in 94 % yield (Scheme 23) [44].



Scheme 22 Oxidative C–H alkynylation of azoles with acetylenes



Scheme 23 Oxidative C–H alkynylation of benzamides with silylacetylene

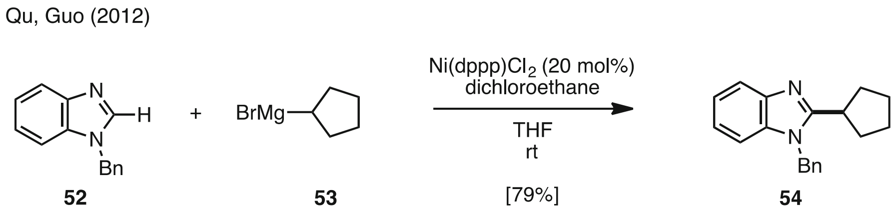
3.3 Oxidative C–H Alkylation

Nickel-catalyzed oxidative C–H alkylation was only recently reported by Qu, Guo, and coworkers in 2012 (Scheme 24) [45]. For example, benzimidazole **52** can be coupled with Grignard reagent **53** using 20 mol% of [1,3′-bis(diphenylphosphino)propane]nickel dichloride [Ni(dppp)Cl_2], and 1,2-dichloroethane as an oxidant in THF to afford the coupling product **54** in 79 % yield. When using these coupling partners, other metal catalysts such as palladium and iron were not effective and only trace amounts of product were obtained.

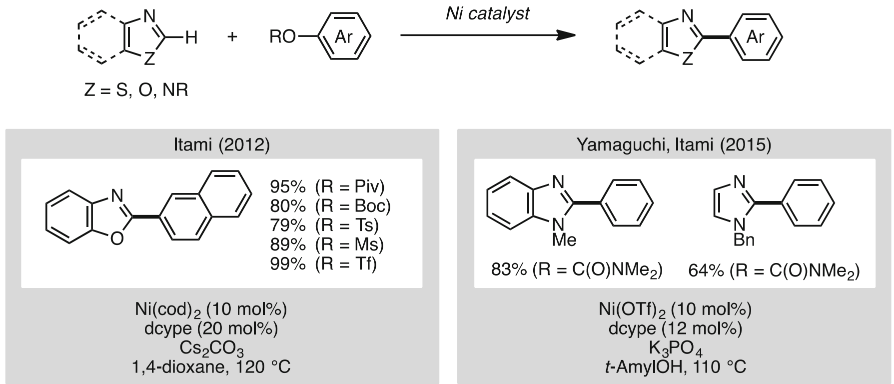
4 Aromatic C–H Functionalization with Unconventional Coupling Partners

4.1 Phenol and Alcohol Derivatives

In 2012, the first nickel-catalyzed C–H/C–O coupling of 1,3-azoles and phenol derivatives was reported by the Itami group (Scheme 25) [46]. In the presence of a Ni(cod)$_2$/dcype catalyst and Cs_2CO_3 as base in 1,4-dioxane, 1,3-azoles and phenol derivatives can be coupled to produce the corresponding 2-aryl-1,3-azoles. Intriguingly, this reaction displays dramatic ligand effects, as other ligands do not deliver coupling products. The Ni(cod)$_2$/dcype catalyst is active for the coupling of other phenol derivatives such as carbamates, carbonates, sulfamates, triflates, tosylates, and mesylates. However, regarding the 1,3-azole coupling partner, it was limited in terms of substrate scope, as imidazoles did not react at all under these reaction conditions. More recently, Itami and Yamaguchi reported a new protocol for the C–H arylation of benzimidazoles and imidazoles with phenol derivatives



Scheme 24 C–H alkylation of benzimidazoles with organomagnesium reagents

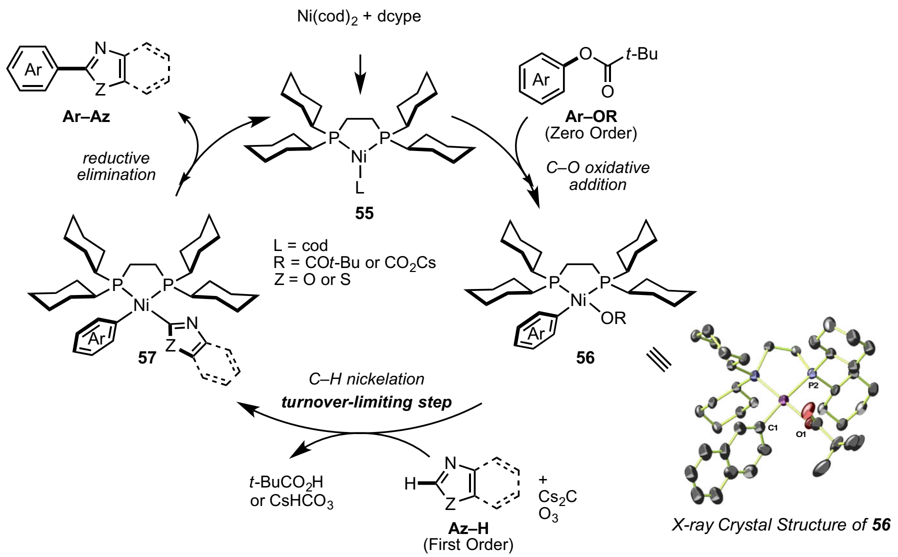


Scheme 25 C–H arylation of azoles with phenol derivatives

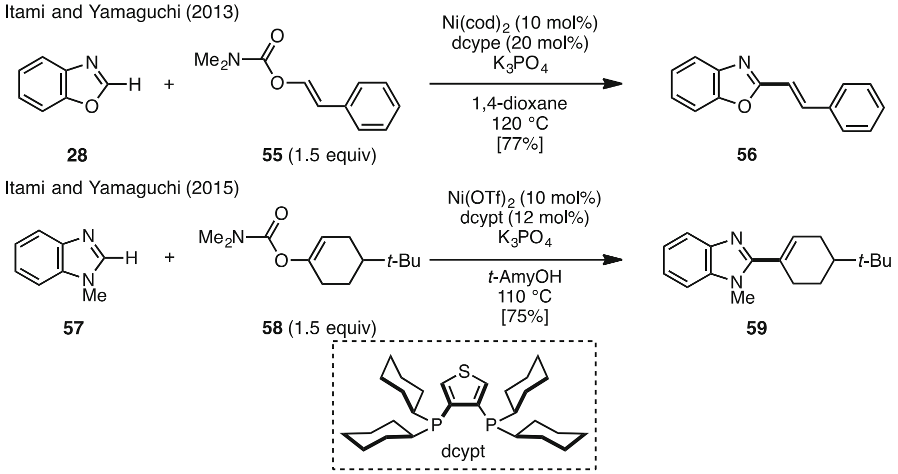
(carbamates) [14]. The catalyst can consist of Ni(II) such as $Ni(OTf)_2$, which is a less expensive complex compared to $Ni(cod)_2$.

Scheme 26 depicts a plausible mechanism based on a Ni(0)/Ni(II) redox catalytic cycle, occurring via (1) oxidative addition of the aromatic C–O bond of Ar–OR to Ni(0) species **55**, (2) C–H nickelation of an azole (Az–H) with Ar–Ni(II)–OR species **56**, and (3) reductive elimination from **57** of the heterobiaryl product (Ar–Az) to regenerate the Ni(0) species. Extensive studies involving the isolation of intermediate **56** and kinetic studies have been reported by Itami, Yamaguchi, and Lei [47]. The isolation of key intermediate **56** resulting from C–O oxidative addition revealed this plausible Ni(0)/Ni(II) redox catalytic cycle. Additionally, kinetic studies and kinetic isotope effect investigations showed that the C–H nickelation is the turnover-limiting step in the catalytic cycle. Furthermore, theoretical calculations were performed to gain further insight regarding the effect of base, especially in the C–H nickelation step [48]. Through these combined experimental and computational studies, a detailed catalytic cycle and the dramatic dcype ligand effect on this reaction were unveiled.

In 2013, Itami and Yamaguchi found that Ni/dcype also catalyzes C–H/C–O alkenylation using enol derivatives (Scheme 27) [14, 49]. Benzoxazole (**28**) was coupled with enol derivative **55** under the nickel catalytic system using the dcype ligand to form alkenyl oxazole **56** in 77 % yield by C–H alkenylation at the C2 position of oxazoles. For the C–H alkenylation of imidazole derivatives, it was



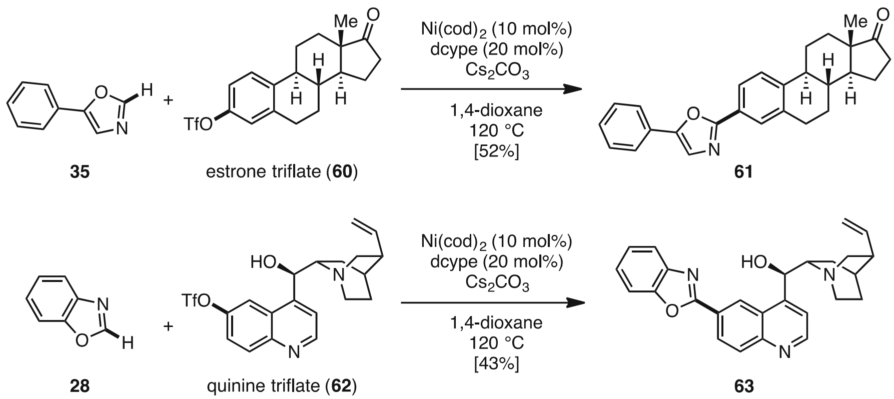
Scheme 26 Plausible catalytic cycle of nickel-catalyzed C–H arylation with phenol derivatives



Scheme 27 C–H alkenylation of azoles with phenol derivatives

found that 3,4-bis(dicyclohexylphosphino)thiophene (dcypt) is an effective ligand, and the coupling reaction of benzimidazole and enol derivative **58** in *t*-AmylOH as a solvent successfully afforded the corresponding coupling product **59** in 75 % yield.

To demonstrate the applicability of this approach, a Ni-catalyzed C–H/C–O coupling to functionalize estrone and quinine was attempted (Scheme 28). The coupling of estrone triflate (**60**) with **35** proceeded smoothly under the standard conditions to afford heteroarylated estrone **61** in 52 % yield. The C–H/C–O coupling of quinine triflate (**62**) with **28** also occurred, giving the quinine–



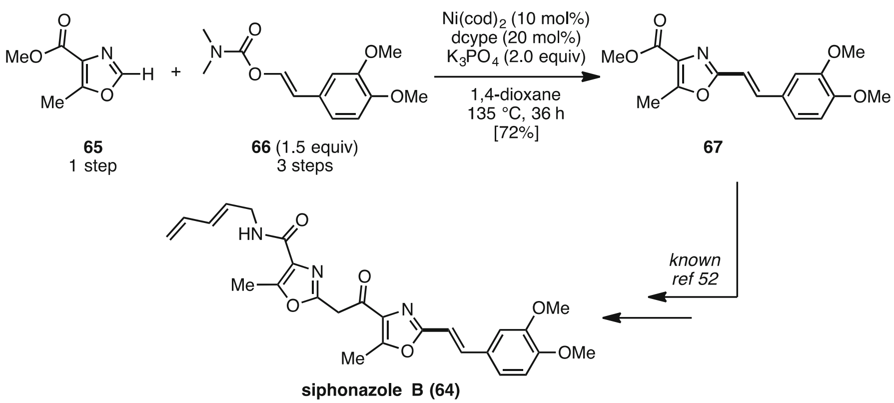
Scheme 28 Late-stage C–O bond heteroarylation of naturally occurring biomolecules by nickel catalysis

benzoxazole hybrid molecule **63**, albeit with somewhat lower efficiency. Notably, the hydroxyl, amine, and olefin functionalities were tolerated under the coupling conditions. This successful application to the functionalization of the quinine structure, which is known to be sensitive under acidic, basic, and redox conditions, highlights the potential of the present nickel catalysis for further development and applications.

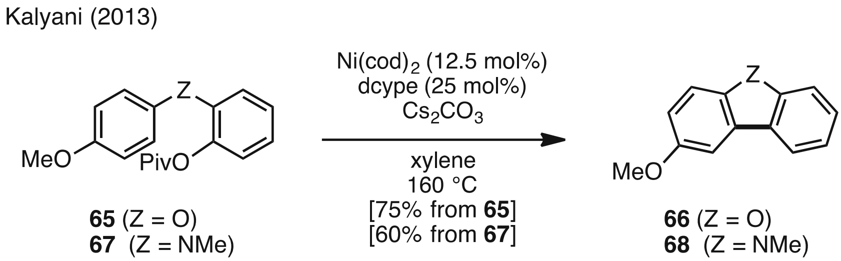
To showcase the utility of this new azole alkenylation method, this reaction was applied to the synthesis of siphonazole B (**64**), a natural product isolated from a metabolite from *Herpetosiphon* sp. (Scheme 29) [50]. Although **64** had already been synthesized by Moody and coworkers [51, 52] as well as Zhang and Ciufolini [53], their syntheses took an excessive number of steps because of a parallel repetition of linear synthetic sequences. To tackle this problem, a convergent synthesis of **64** through nickel-catalyzed C–H alkenylation was designed. Although **67** was a useful intermediate in the previous synthesis, this had required seven linear steps. Thus, this was retrosynthetically divided into oxazole **65** and alkenyl enol **66**. The C–H/C–O alkenylation of **65** and **66** delivered **67** in good yield over four linear steps, thus accomplishing the formal synthesis of siphonazole B (**64**).

Meanwhile, an intramolecular C–H/C–O coupling with the same nickel catalyst was also reported in 2013 by the Kalyani group (Scheme 30) [54]. For example, pivalate **65** was subjected to reaction conditions similar to Itami's group, but in xylene instead of dioxane, to give benzofuran **66** in 75 % yield. This benzofuran synthesis could be expanded to the synthesis of carbazoles (Z = NMe, 60 % yield). This result is the first example of nickel-catalyzed C–H arylation of simple arenes using C–O electrophiles.

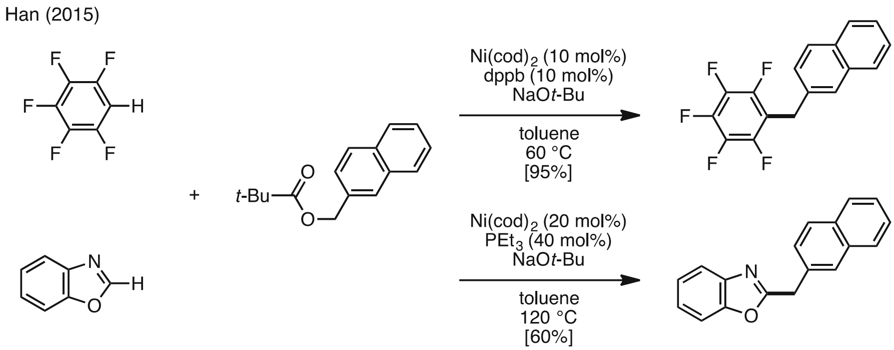
Recently, Han and coworkers reported the nickel-catalyzed C–H alkylations of azoles/pentafluorobenzene with benzylic alcohol derivatives (Scheme 31) [55]. Using a $Ni(cod)_2$/dppb catalytic system and NaO*t*-Bu as a base, 1,3-azoles and pentafluorobenzene could be alkylated with pivalates at the benzyl position to give the corresponding products in good to excellent yields. Although these reactions seem to proceed via nucleophilic substitution, they did not proceed without nickel catalyst.



Scheme 29 Synthesis of siphonazole B



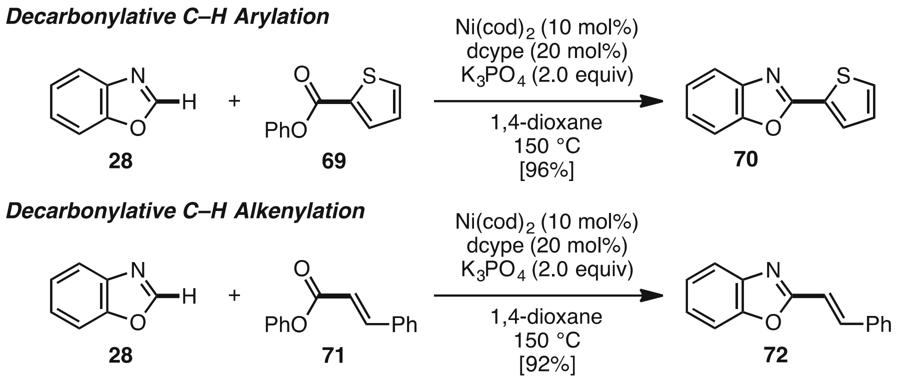
Scheme 30 Dibenzofuran and carbazole synthesis through intramolecular C–H/C–O coupling



Scheme 31 C–H alkylation of fluorobenzenes and azoles with benzyl alcohol derivatives

4.2 Esters and Carboxylic Acids

In 2012, the group of Itami discovered the first nickel-catalyzed decarbonylative C–H biaryl coupling of azoles and aryl esters (Scheme 32) [56]. Under a catalytic system similar to the aforementioned nickel-catalyzed Ar–H/Ar–O coupling (see Scheme 24), decarbonylative C–H coupling of benzoxazole (**28**) and phenyl



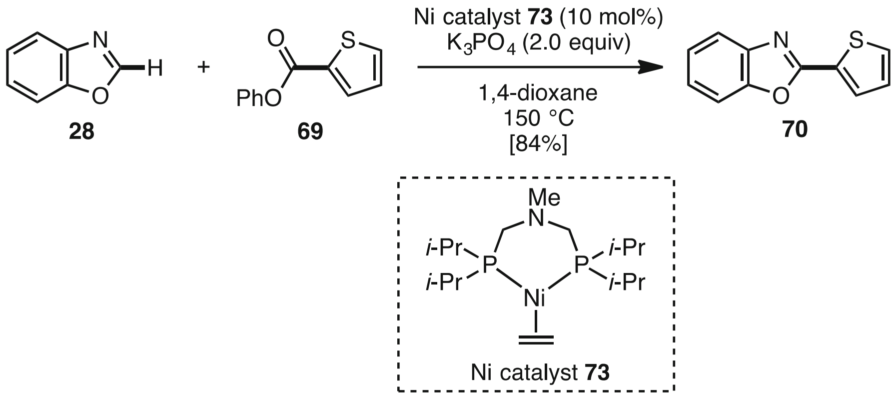
Scheme 32 Decarbonylative C–H arylation and alkenylation of azoles

thiophenecarboxylate (**69**) proceeds smoothly to furnish the corresponding coupling product **70** in 96 % yield. Various esters, particularly heteroaromatic esters such as furans, thiophenes, thiazoles, pyridines, and quinolines, can be used in this reaction to give the corresponding coupling products. Additionally, these reaction conditions can be applied to the decarbonylative C–H alkenylation of azoles and α,β-unsaturated phenyl esters such as **71** [49].

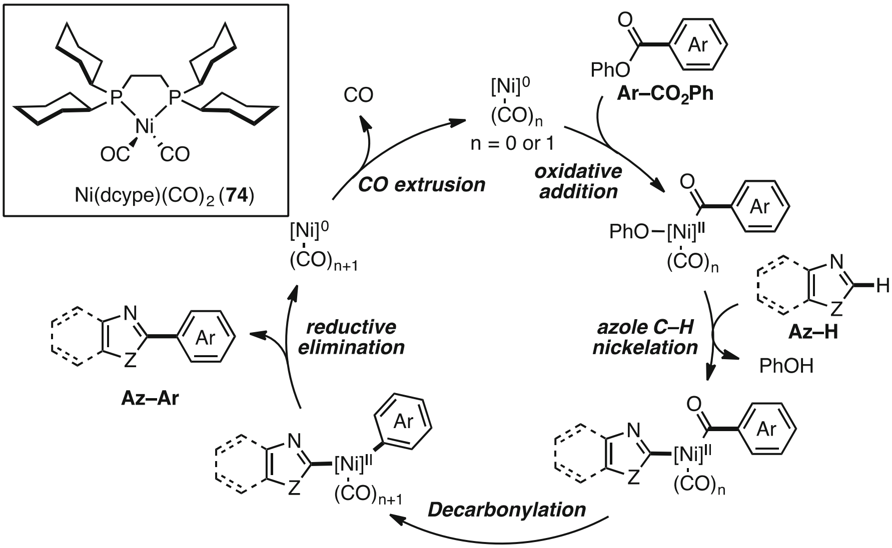
Thereafter, Gade and coworkers also demonstrated the same reaction using a unique nickel catalyst (Scheme 33) [57]. They prepared nickel complex **73** from [$NiCl_2Py_4$] and bis(diisopropylphosphinomethyl)amine, and coupling of **28** and **69** was conducted in the presence of 10 mol% **73** to afford the coupling product **70** in 84 % yield.

A plausible reaction mechanism of the decarbonylative C–H functionalization is shown in Scheme 34. The reaction might proceed through Ni^0/Ni^{II} redox catalysis involving (1) oxidative addition of the phenyl ester C–O bond onto Ni^0; (2) C–H nickelation of azole (Het–H) with Ar–$Ni^{II}(CO)_n$–OPh to generate Ar–$Ni^{II}(CO)_n$–Het; (3) CO migration (decarbonylation) onto the nickel center to produce an Ar–$Ni^{II}(CO)_{n+1}$–Het species ($n = 0$ or 1) [58]; and (4) reductive elimination to release the coupling product (Het–Ar) and to generate a $Ni^0(CO)_{n+1}$ species. Although a seemingly inactive nickel dicarbonyl complex **74** would be produced after two turnovers, the active Ni^0 catalyst could be regenerated by thermal extrusion of CO from **74**. After Itami's and Gade's reports, the group of Houk [59] and Lu [60] independently reported the mechanistic studies of this decarbonylative coupling as well as Ar–H/Ar–O coupling (see Scheme 26) by density functional theory (DFT) calculations.

This newly developed decarbonylative C–H functionalization is useful in complex natural product synthesis. For example, Itami and Yamaguchi achieved the synthesis of muscoride A (**74**), a natural product with antibacterial activity (Scheme 35) [56, 61]. Two azole esters, **75** and (−)-**76**, were coupled under Ni/dcype catalysis to furnish the corresponding coupling product (−)-**77** in 39 % yield. As the conversion of (−)-**77** into (−)-**74** had been previously described [62], a formal synthesis of (−)-muscoride A (**74**) was completed. If the synthesis was planned and executed with



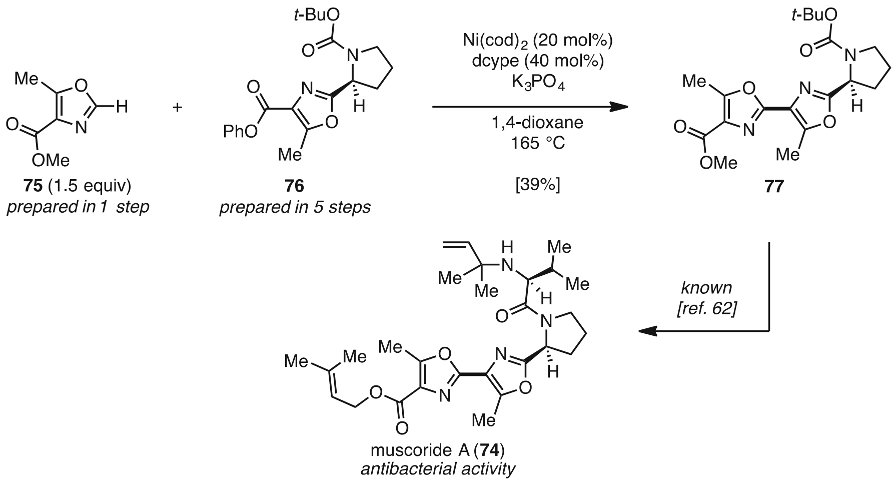
Scheme 33 Decarbonylative C–H arylation by an alternative nickel catalyst



Scheme 34 Plausible reaction mechanism of decarbonylative C–H functionalization

typical cross-coupling substrates (aryl halides and organometallic reagents) instead, it would have become much less efficient, with many added steps.

Very recently, Yamaguchi and Itami also applied their decarbonylative C–H arylation to the formal synthesis of thiopeptide antibiotics GE2270s (Scheme 36) [63]. Two azole esters **78** and **79**, which were prepared in five steps and three steps respectively from commercially available compounds, were coupled under modified conditions to afford the corresponding coupling product **80** in 49 % yield. Next, coupling product **80** was reacted with thiazolyl acrylic acid **81** in *o*-dichlorobenzene at 180 °C via a decarboxylative Diels–Alder reaction to afford the corresponding

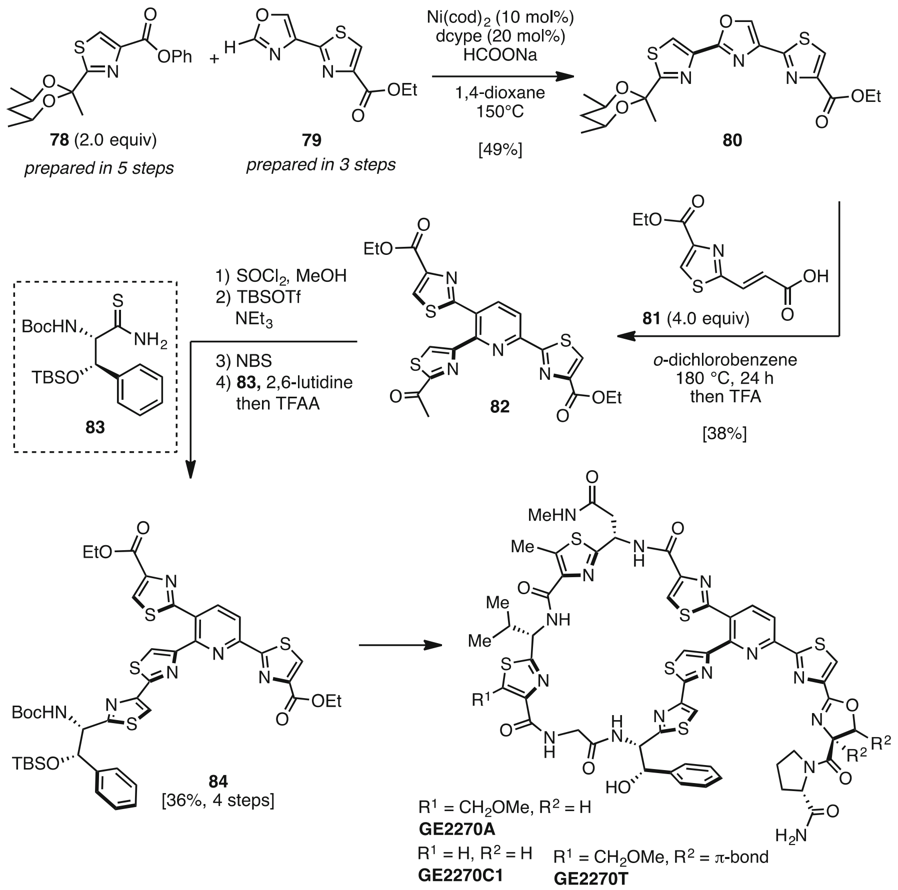


Scheme 35 Synthesis of muscoride A via decarbonylative C–H arylation

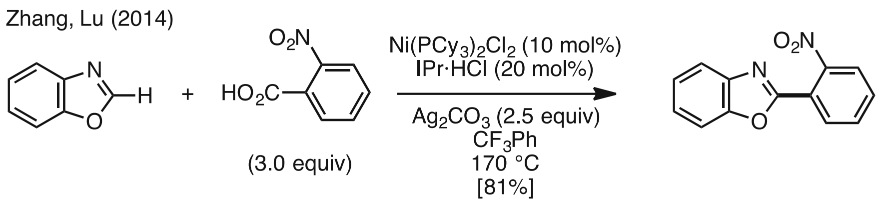
trithiazolylpyridine. Subsequently, removal of the acetal group by trifluoroacetic acid (TFA) afforded **82** in 38 % yield over two steps. The key trithiazolylpyridine **82** can be converted to thiopeptide antibiotic precursors. Compound **82** was treated with *tert*-butyldimethylsilyl trifluoromethanesulfonate (TBSOTf) and NEt_3 followed by bromination with *N*-bromosuccinimide (NBS) to afford brominated products. These products were treated with thioamide **83** followed by trifluoroacetic anhydride (TFAA) to afford **84** in 36 % yield (four steps). Since the conversion of **84** to GE2270s had been described previously by Nicolaou and coworkers [64, 65], the formal syntheses of GE2270s were accomplished through intermediate **82**.

Zhang and coworkers reported a nickel-catalyzed decarboxylative coupling of 1,3-azoles and carboxylic acid derivatives (Scheme 37) [66]. The catalyst $Ni(PCy_3)_2Cl_2$ in conjunction with IPr·HCl was effective in increasing yields. Although this catalytic transformation was already reported with other metals such as palladium, copper, iron, rhodium, and ruthenium, they were the first to investigate this type of reaction using nickel catalyst. Although electron-deficient substituents such as nitro or fluoro groups were required at the *ortho* position of arenecarboxylic acids, 2-arylazoles can be obtained from arenecarboxylic acids as the coupling partner.

Ge and coworkers also reported a nickel-catalyzed decarboxylative coupling using α-oxoglyoxylic acids as coupling partners (Scheme 38) [67]. Although a similar palladium-catalyzed C–H acylation using α-oxoglyoxylic acids had already been reported, it was the first time in which C–H acylation proceeded by nickel catalysis. For example, benzoxazole (**28**) and phenyloxoglyoxylic acid (**85**) were heated with a catalytic amount of $Ni(ClO_4)_2{\cdot}6H_2O$ and stoichiometric Ag_2CO_3 in benzene to afford acylation product **86** in 85 % yield.



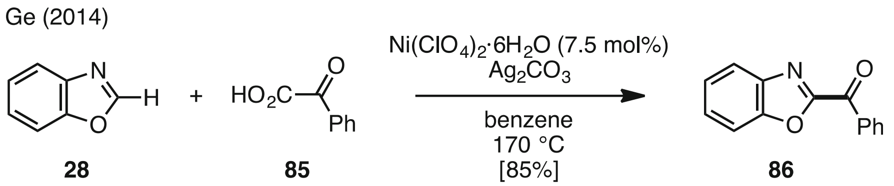
Scheme 36 Formal syntheses of GE2270s



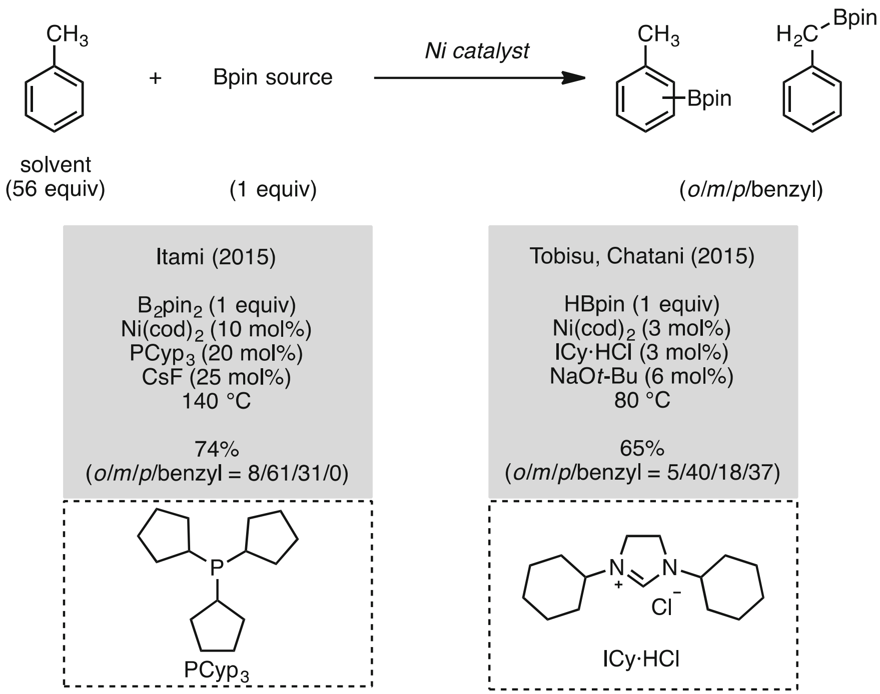
Scheme 37 Decarbonylative coupling of 1,3-azoles and arenecarboxylic acids

5 Introduction of Heteroatoms at Aromatic C–H Bonds

Examples of introducing heteroatoms at aromatic C–H bonds by nickel catalysts are still rare, and only several studies have been reported for aromatic C–H borylation and thiolation. In 2015, the Itami group and the Tobisu–Chatani group



Scheme 38 Decarboxylative C–H acylation of 1,3-azoles and α-oxoglyoxylic acids

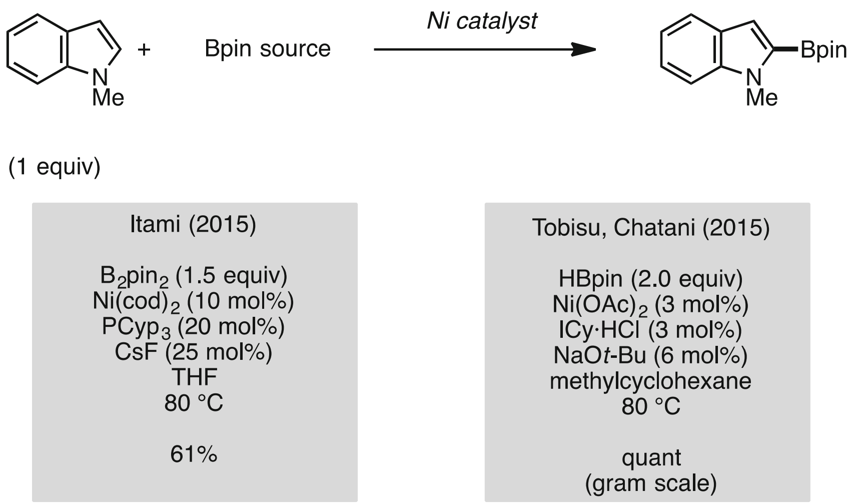


Scheme 39 C–H borylation of arenes

independently reported the C–H borylation of (hetero)aromatic rings by nickel catalysis (Scheme 39) [68, 69]. The Itami group used a $Ni(cod)_2$/tricyclopentylphosphine ($PCyp_3$)/CsF catalytic system and bis(pinacolato)diboron (B_2pin_2) as the borylation agent, whereas Tobisu and Chatani used a $Ni(cod)_2$/ICy·HCl/NaO*t*-Bu catalytic system and pinacolborane (HBpin) as the borylation agent. Under both conditions, toluene was preferentially borylated at the *meta* position, along with a mixture of *ortho*, *para*, and benzylic borylation products.

These reaction conditions were not only applicable when using excess amounts of simple arenes but also when using stoichiometric amounts of heteroaromatics such as indoles (Scheme 40) [68, 69].

In contrast, the groups of Lu, Shi, and Kambe independently reported a nickel-catalyzed C–H thiolation of benzamide derivatives containing a chelation-assisted



Scheme 40 C–H borylation of indoles

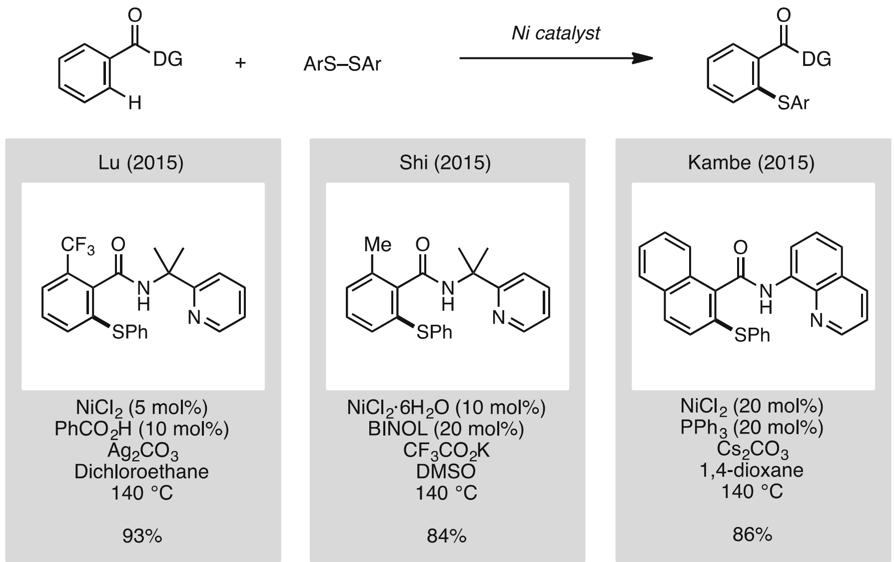
group in 2015 (Scheme 41) [70–72]. Although the chelation-assisted group differed (2-pyridyldimethylamine by Lu and Shi, and 8-aminoquinoline by Kambe), they all used disulfides for the thiolating reagent.

6 Hydroarylation-Type C–H Functionalization

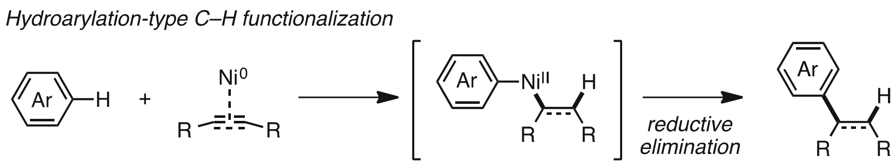
The addition of a (hetero)arene C–H bond onto alkynes or alkenes under transition metal catalysis to generate alkenyl or alkyl arenes can be classified as a hydroarylation-type C–H functionalization (Scheme 42). Although hydroarylations using other metal catalysts have been reported, this section focuses on nickel-catalyzed hydroarylation-type C–H functionalization. Mechanistically, these reactions typically start with the coordination of an alkyne/alkene to a Ni(0) species. The complex is reacted with a (hetero)aromatic C–H bond via oxidative addition, followed by hydronickelation forming an Ar–Ni(II)–alkenyl (alkyl) intermediate, which then undergoes reductive elimination to afford hydro(hetero)arylation products. Since an excellent review of this reaction type by nickel catalysis has already been reported by Nakao [6], details are omitted and only reaction types are classified here.

6.1 Hydroarylation of Alkynes

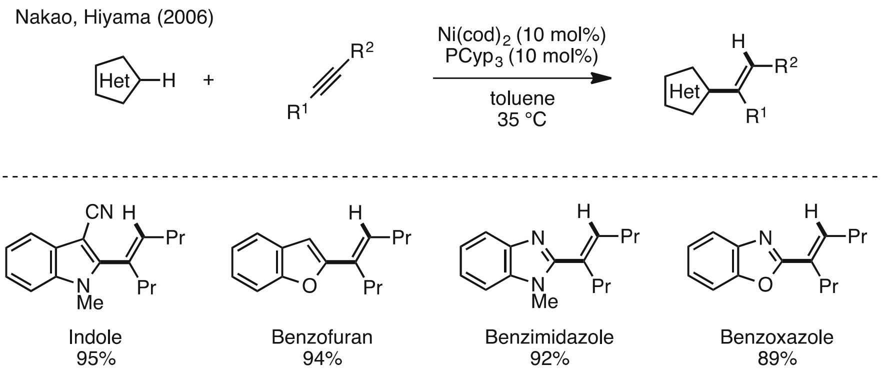
In 2006, Nakao, Hiyama, and coworkers reported a nickel-catalyzed hydroarylation of alkynes with heteroarenes (Scheme 43) [73]. Various heteroarenes such as indoles, benzofurans, benzimidazoles, and benzoxazoles reacted with alkynes in the presence of $Ni(cod)_2/PCyp_3$ in toluene under mild conditions to give the



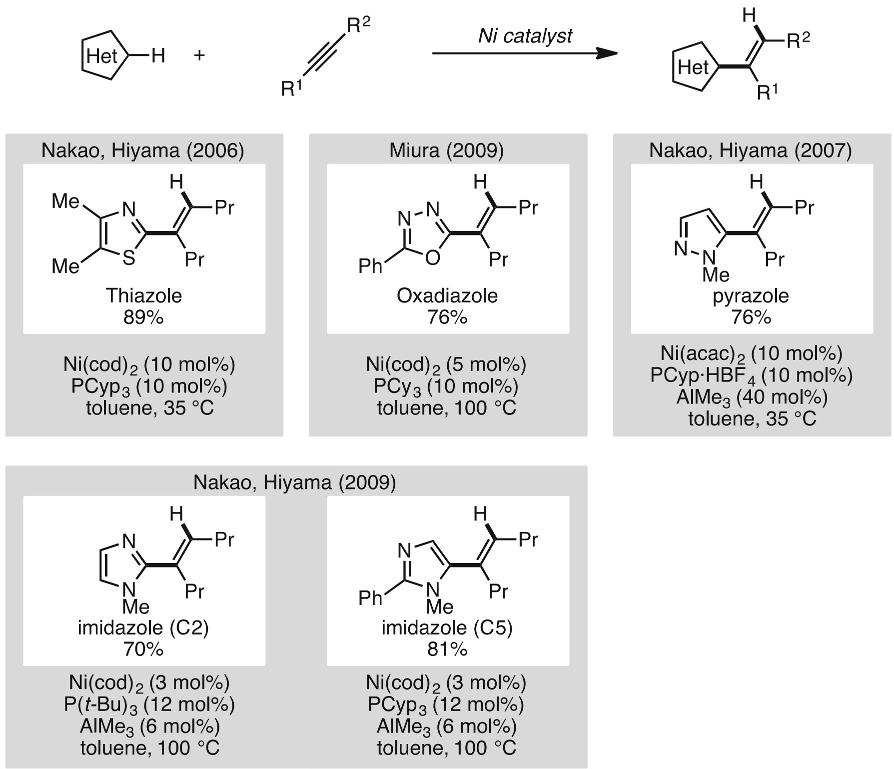
Scheme 41 C–H thiolation of benzamides



Scheme 42 Hydroarylation-type C–H functionalization



Scheme 43 Hydroarylation of alkynes using heteroarenes

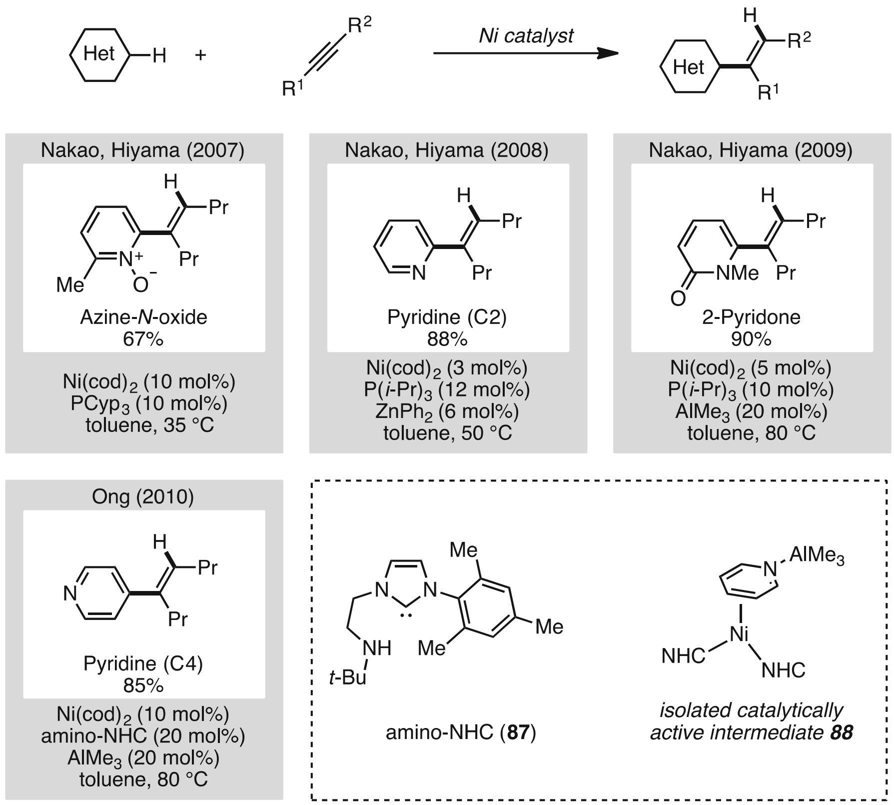


Scheme 44 Hydroarylation of alkynes with 5-membered heteroarenes

corresponding heteroaryl-substituted ethenes with high chemo-, stereo-, and regioselectivity.

This nickel-catalyzed hydroarylation is applicable not only for benzoheteroarenes, but also for five-membered heteroarenes (Scheme 44). For example, Nakao and Hiyama demonstrated the hydroarylation of thiazoles using the same nickel catalyst [73], and the Miura group reported the hydroarylation of oxadiazoles by using a $Ni(cod)_2/PCyp_3$ catalytic system [74]. Additionally, Nakao and Hiyama discovered that $AlMe_3$ is an effective additive for pyrazoles and imidazoles [75, 76]. These reactions typically proceeded selectively at the C2 position on heteroarenes. However, when imidazoles bearing a C2 substituent were used, hydroarylation proceeded at the C5 position of imidazoles. When using dialkylethynes, the reaction proceeded with *syn* addition, whereas diarylethynes led to *anti*-addition products.

The first example of nickel-catalyzed hydroarylation of electron-deficient heteroaromatics was reported by Nakao and Hiyama in 2007 (Scheme 45) [77]. Alkynes were reacted with azine *N*-oxides under nickel catalysis to afford alkenyl products in good yields. In 2008, they discovered that the addition of Lewis acids such as $ZnMe_2$ or $ZnPh_2$ instead of using *N*-oxides is effective, affording the corresponding hydroarylation products regioselectively at the C2 position on



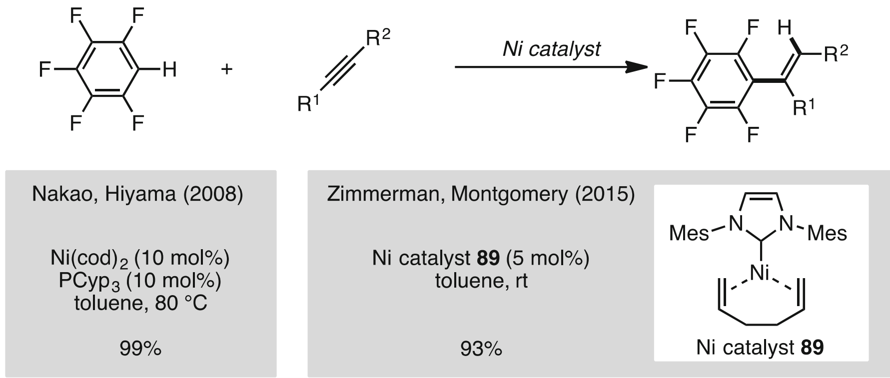
Scheme 45 Hydroarylation of alkynes with electron-deficient heteroaromatics

pyridines [78]. In 2009, they also applied modified catalytic conditions to the alkenylation of pyridone derivatives [79]. In contrast, Ong and coworkers reported a nickel-catalyzed hydroarylation of alkynes and pyridines at the C4 position [80]. They assumed that the complexation of amino-NHC (**87**) and $AlMe_3$ such as **88** activated a C–H bond on pyridine at the C4 position.

This nickel catalysis enables the hydroarylation of alkynes with fluoroarenes (Scheme 46). In 2008, Nakao and Hiyama demonstrated that pentafluorobenzene can react with alkynes using a $Ni(cod)_2/PCyp_3$ catalyst [81]. Recently, Zimmerman and Montgomery found that 1,5-cyclooctadiene (cod) plays an inhibitory role in catalytic efficiency in nickel-catalyzed reactions, and therefore they used precatalyst **89** involving 1,5-hexadiene instead of cod [82]. With this modification, the hydroarylation of alkynes with fluoroarenes smoothly proceeded at room temperature to afford the corresponding alkenylated products in good yields.

6.2 Hydroarylation of Alkenes

Hydroarylation is applicable not only for alkynes but also for alkenes (Scheme 47). In 2010, Nakao and Hiyama reported a nickel-catalyzed hydroheteroarylation of



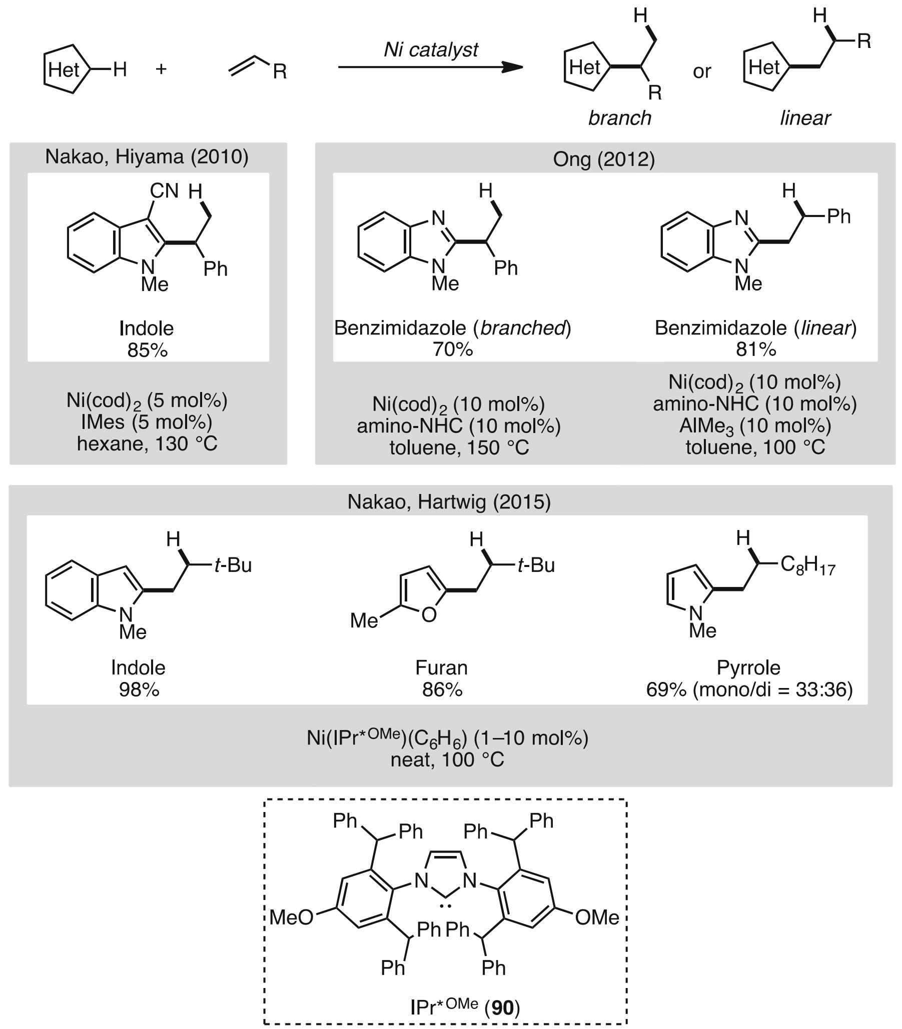
Scheme 46 Hydroarylation of alkynes with pentafluorobenzenes

alkenes with benzo-annulated heteroarenes such as indoles using a $Ni(cod)_2$/IMes catalytic system [83]. Ong and coworkers reported a similar reaction using $Ni(cod)_2$ and an amino-NHC ligand (**87**) [84, 85]. In their investigation, it was found that the regioselectivity switches when using $AlMe_3$ as the additive. The hydroarylation normally affords branched products, whereas linear adducts (anti-Markovnikov products) are obtained when using $AlMe_3$. Recently, collaborative research by the groups of Nakao and Hartwig also demonstrated an anti-Markovnikov hydroheteroarylation of alkenes catalyzed by a hindered nickel NHC (**90**) system [86]. This reaction proceeded with high anti-Markovnikov selectivity and showed broad substrate scope, as it was applicable for unactivated alkenes, internal alkenes, and cyclic alkenes as the alkene component and indoles, pyrroles, benzofurans, and furans as the heteroarene component.

For the hydroarylation of alkenes with electron-deficient heteroaromatics such as pyridines, Nakao and Hiyama achieved regioselective hydroarylation when using a bulky Lewis acid additive, methylaluminum bis(2,6-di-*tert*-butyl-4-methylphenoxide) (MAD) (Scheme 48) [87]. In this reaction, alkenes bearing aromatics afforded branched products, whereas alkenes bearing alkyl chains gave linear products.

In 2014, the Hartwig group also demonstrated a nickel-catalyzed hydroarylation of electron-deficient aromatics (Scheme 49) [88]. Although only 1,3-bis(trifluoromethyl)benzene worked well, catalytic $Ni(IPr)_2$/NaO*t*-Bu gave alkylated benzene derivatives in good to moderate yields.

Typically, the hydroarylation of alkynes and alkenes is applicable only for heteroaromatics and specific arenes (see Sect. 6). In the case of simple arenes such as benzoic acid derivatives and anilines, utilization of chelation-assisted groups is required for enhanced reactivity. In 2011, Chatani and coworkers attempted a nickel-catalyzed hydroarylation of alkynes with benzamides containing a 2-pyridylaminomethane moiety and obtained isoquinoline derivatives via oxidative cyclization (annulation) (Scheme 50) [89]. Although similar types of reactions have already been reported using noble metals such as ruthenium and rhodium, this was the first example of nickel-catalyzed *ortho* C–H bond activation and hydroarylation of alkynes with simple arenes. Ackermann and coworkers also

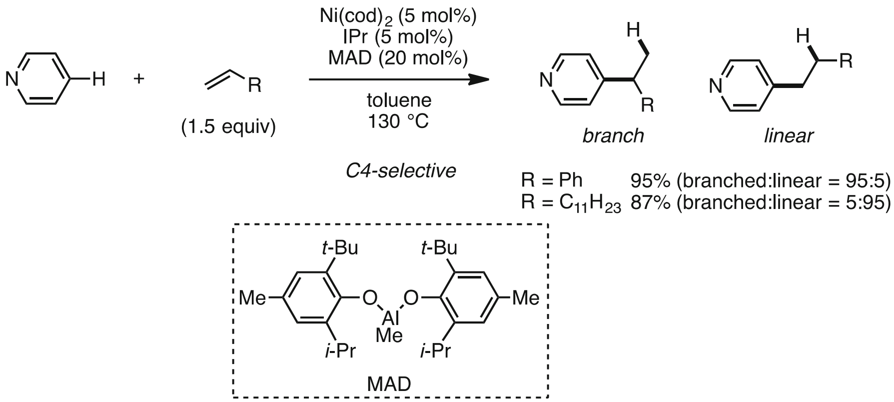


Scheme 47 Hydroarylation of alkenes with electron-rich heteroaromatics

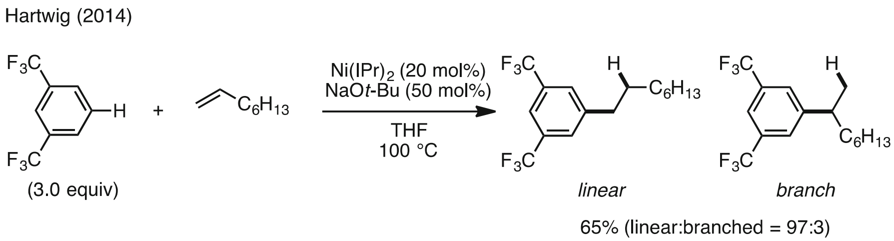
reported a similar type of reaction involving a nickel-catalyzed annulation of anilines and alkynes to form substituted indoles [90].

7 Summary

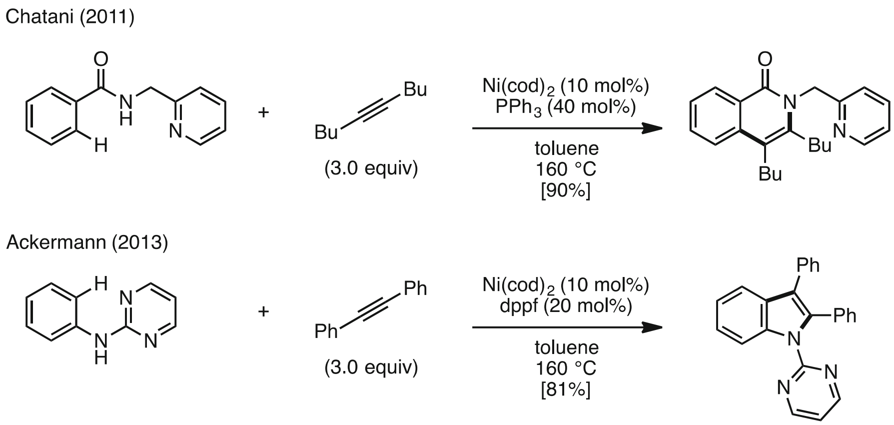
The field of nickel-catalyzed aromatic C–H functionalization has evolved significantly in the past decade. Such reactions are not only advantageous because of the low cost associated with nickel catalysts but they also enable unique transformations on aromatic starting materials for conversion into new organic materials, natural



Scheme 48 Hydroarylation of alkenes with pyridines



Scheme 49 Hydroarylation of alkenes with electron-deficient arenes



Scheme 50 Alkyne annulation via *ortho* C–H bond activation

products, and pharmaceuticals. Additionally, while this review focused on aromatic C–H functionalization, nickel-catalyzed sp^3 C–H functionalization with haloarenes [91–97], oxidative coupling [98–102], heteroatom formation [103–107], hydrocarboxylation and hydroalkylation of alkynes and alkenes [108–111], and annulation [112] also appeared recently. With continued development of novel catalysts [113–117], nickel-catalyzed C–H functionalization is becoming a broadly applicable and reliable synthetic method for next-generation organic synthesis.

References

1. Tasker SZ, Standley EA, Jamison TF (2014) Nature 509:299–309
2. Yamaguchi J, Muto K, Itami K (2013) Eur J Org Chem 19–30
3. Cai X-H, Xie B (2015) ARKIVOC 184–211
4. Hyster TK (2015) Catal Lett 145:458–467
5. Khan MS, Haque A, Al-Suti MK, Raithby PR (2015) J Organomet Chem 793:114–133
6. Nakao Y (2011) Chem Rec 11:242–251
7. Castro LCM, Chatani N (2015) Chem Lett 44:410–421
8. Klelman JP, Dubeck M (1963) J Am Chem Soc 85:1544–1545
9. Liang L-C, Cheng P-S, Huang Y-L (2006) J Am Chem Soc 128:15562–15563
10. Canivet J, Yamaguchi J, Ban I, Itami K (2009) Org Lett 11:1733–1736
11. Hachiya H, Hirano K, Satoh T, Miura M (2009) Org Lett 11:1737–1740
12. Yamamoto T, Muto K, Komiyama M, Canivet J, Yamaguchi J, Itami K (2011) Chem Eur J 17:10113–10122
13. Iaroshenko V, Ali I, Mkrtchyan S, Semeniuchenko V, Ostrovskyi D, Langer P (2012) Synlett 23:2603–2608
14. Muto K, Hatakeyama T, Yamaguchi J, Itami K (2015) Chem Sci 6:6792–6798
15. Watanabe K, Tanaka T, Kondo S (1994) Jpn Kokai Tokkyo Koho 329647
16. Razavi H, Palaninathan SK, Powers ET, Wiseman RL, Purkey HE, Mohamedmohaideen NN, Deechongkit S, Chiang KP, Dendle MTA, Sacchettini JC, Kelly JW (2003) Angew Chem Int Ed 42:2758–2761
17. Domînguez XA, de La Fuente G, Gonzal ez AG, Rein M, Timon I (1998) Heterocycles 27:35–38
18. Kobayashi O, Uraguchi D, Yamakawa T (2009) Org Lett 11:2679–2682
19. Yokota A, Aihara Y, Chatani N (2014) J Org Chem 79:11922–11932
20. Matsuyama N, Hirano K, Satoh T, Miura M (2009) Org Lett 11:4156–4159
21. Liu YJ, Liu YH, Yan SY, Shi BF (2015) Chem Commun 51:6388–6391
22. Yi J, Yang L, Xia C, Li F (2015) J Org Chem 80:6213–6221
23. Landge VG, Shewale CH, Jaiswal G, Sahoo MK, Midya SP, Balaraman E (2016) Catal Sci Technol 6:1946–1951
24. Cho H, Iwama Y, Okano K, Tokuyama H (2014) Chem Pharm Bull 62:354–363
25. Vechorkin O, Proust V, Hu X (2010) Angew Chem Int Ed 49:3061–3064
26. Yao T, Hirano K, Satoh T, Miura M (2010) Chem Eur J 16:12307–12311
27. Yao T, Hirano K, Satoh T, Miura M (2012) Angew Chem Int Ed 51:775–779
28. Liu C, Liu D, Zhang W, Zhou L, Lei A (2013) Org Lett 15:6166–6169
29. Aihara Y, Chatani N (2013) J Am Chem Soc 135:5308–5311
30. Aihara Y, Wuelbern J, Chatani N (2015) Bull Chem Soc Jpn 88:438–446
31. Uemura T, Yamaguchi M, Chatani N (2016) Angew Chem Int Ed 55:3162–3165
32. Barsu N, Kalsi D, Sundararaju B (2015) Chem Eur J 21:9364–9368
33. Cong X, Li Y, Wei Y, Zeng X (2014) Org Lett 16:3926–3929
34. Song W, Lackner S, Ackermann L (2014) Angew Chem Int Ed 53:2477–2480
35. Ruan Z, Lackner S, Ackermann L (2016) Angew Chem Int Ed 55:3153–3157
36. Wu X, Zhao Y, Ge H (2015) J Am Chem Soc 137:4924–4927
37. Hachiya H, Hirano K, Satoh T, Miura M (2010) Angew Chem Int Ed 49:2202–2205
38. Hachiya H, Hirano K, Satoh T, Miura M (2010) Chem Cat Chem 2:1403–1406
39. Qu G-R, Xin P-Y, Niu H-Y, Wang D-C, Ding R-F, Guo H-M (2011) Chem Commun 47:11140–11142
40. Tobisu M, Hyodo I, Chatani N (2009) J Am Chem Soc 131:12070–12071

41. Hyodo I, Tobisu M, Chatani N (2012) Chem Asian J 7:1357–1365
42. Hyodo I, Tobisu M, Chatani N (2012) Chem Commun 48:308–310
43. Matsuyama N, Kitahara M, Hirano K, Satoh T, Miura M (2010) Org Lett 12:2358–2361
44. Liu Y-H, Liu Y-J, Yan S-Y, Shi B-F (2015) Chem Commun 51:11650–11653
45. Xin P-Y, Niu H-Y, Qu G-R, Ding R-F, Guo H-M (2012) Chem Commun 48:6717–6719
46. Muto K, Yamaguchi J, Itami K (2012) J Am Chem Soc 134:169–172
47. Muto K, Yamaguchi J, Lei A, Itami K (2013) J Am Chem Soc 135:16384–16387
48. Xu H, Muto K, Yamaguchi J, Zhao C, Itami K, Musaev DG (2014) J Am Chem Soc 136:14834–14844
49. Meng L, Kamada Y, Muto K, Yamaguchi J, Itami K (2013) Angew Chem Int Ed 52:10048–10051
50. Nett M, Erol Ö, Kehraus S, Köck M, Krick A, Eguereva E, Neu E, König GM (2006) Angew Chem Int Ed 45:3863–3867
51. Linder JR, Moody CJ (2007) Chem Commun 1508–1509
52. Linder JR, Blake AJ, Moody CJ (2008) Org Biomol Chem 6:3908–3916
53. Zhang J, Ciufolini MA (2009) Org Lett 11:2389–2392
54. Wang J, Ferguson DM, Kalyani D (2013) Tetrahedron 69:5780–5790
55. Xiao J, Chen T, Han L-B (2015) Org Lett 17:812–815
56. Amaike K, Muto K, Yamaguchi J, Itami K (2012) J Am Chem Soc 134:13573–13576
57. Kruckenberg A, Wadepohl H, Gade LH (2013) Organometallics 32:5153–5170
58. YamamotoT, Ishizu J, Kohara T, Komiya S, Yamamoto A (1980) J Am Chem Soc 102:3758–3764
59. Hong X, Liang Y, Houk KN (2014) J Am Chem Soc 136:2017–2025
60. Lu Q, Yu H, Fu Y (2014) J Am Chem Soc 136:8252–8260
61. Nagatsu A, Kajitani H, Sakakibara J (1995) Tetrahderon Lett 36:4097–4100
62. Wipf P, Venkatraman S (1996) J Org Chem 61:6517–6522
63. Amaike K, Itami K, Yamaguchi J (2016) Chem Eur J 22:4384–4388
64. Nicolaou KC, Zou B, Dethe DH, Li DB, Chen DY-K (2006) Angew Chem Int Ed 45:7786–7792
65. Nicolaou KC, Dethe DH, Leung GY, Zou B, Chen DY-K (2008) Chem Asian J 3:413–429
66. Yang K, Wang P, Zhang C, Kadi AA, Fun H-K, Zhang Y, Lu H (2014) Eur J Org Chem 7586–7589
67. Yang K, Zhang C, Wang P, Zhang Y, Ge H (2014) Chem Eur J 20:7241–7244
68. Zhang H, Hagihara S, Itami K (2015) Chem Lett 44:779–781
69. Furukawa T, Tobisu M, Chatani N (2015) Chem Commun 51:6508–6511
70. Yang K, Wang Y, Chen X, Kadi AA, Fun H-K, Sun H, Zhang Y, Lu H (2015) Chem Commun 51:3582–3585
71. Yan S-Y, Liu Y-J, Liu B, Liu Y-H, Shi B-F (2015) Chem Commun 51:4069–4072
72. Reddy VP, Qiu R, Iwasaki T, Kambe N (2015) Org Biomol Chem 13:6803–6813
73. Nakao Y, Kanyiva KS, Oda S, Hiyama T (2006) J Am Chem Soc 128:8146–8147
74. Mukai T, Hirano K, Satoh T, Miura M (2009) J Org Chem 74:6410–6413
75. Kanyiva KS, Nakao Y, Hiyama T (2007) Heterocycles 72:677–680
76. Kanyiva KS, Löbermann F, Nakao Y, Hiyama T (2009) Tetrahedron Lett 50:3463–3466
77. Kanyiva KS, Nakao Y, Hiyama T (2007) Angew Chem Int Ed 46:8872–8874
78. Nakao Y, Kanyiva KS, Hiyama T (2008) J Am Chem Soc 130:2448–2449
79. Nakao Y, Idei H, Kanyiva KS, Hiyama T (2009) J Am Chem Soc 131:15996–15997
80. Tsai CC, Shih WC, Fang CH, Li CY, Ong TG, Yap GPA (2010) J Am Chem Soc 132:11887–11889
81. Nakao Y, Kashihara N, Kanyiva KS, Hiyama T (2008) J Am Chem Soc 130:16170–16171
82. Nett AJ, Zhao W, Zimmerman PM, Montgomery J (2015) J Am Chem Soc 137:7636–7639
83. Nakao Y, Kashihara N, Kanyiva KS, Hiyama T (2010) Angew Chem Int Ed 49:4451–4454
84. Shih WC, Chen WC, Lai YC, Yu MS, Ho JJ, Yap GPA, Ong TG (2012) Org Lett 14:2046–2049
85. Lee WC, Shih WC, Wang TH, Liu Y, Yap GPA, Ong TG (2015) Tetrahedron 71:4460–4464
86. Schramm Y, Takeuchi M, Semba K, Nakao Y, Hartwig JF (2015) J Am Chem Soc 137:12215–12218
87. Nakao Y, Yamada Y, Kashihara N, Hiyama T (2010) J Am Chem Soc 132:13666–13668
88. Bair JS, Schramm Y, Sergeev AG, Clot E, Eisenstein O, Hartwig JF (2014) J Am Chem Soc 136:13098–13101
89. Shiota H, Ano Y, Aihara Y, Fukumoto Y, Chatani N (2011) J Am Chem Soc 133:14952–14955
90. Song W, Ackermann L (2013) Chem Commun 49:6638–6640
91. Fernández-Salas JA, Marelli E, Nolan SP (2015) Chem Sci 6:4973–4977
92. Aihara Y, Chatani N (2014) J Am Chem Soc 136:898–901
93. Li M, Dong J, Huang X, Li K, Wu Q, Song F, You J (2014) Chem Commun 50:3944–3946

94. Iyanaga M, Aihara Y, Chatani N (2014) J Org Chem 79:11933–11939
95. Wang X, Zhu L, Chen S, Xu X, Au C-T, Qiu R (2015) Org Lett 17:5228–5231
96. Liu Y-J, Zhang Z-Z, Yan S-Y, Liu Y-H, Shi B-F (2015) Chem Commun 51:7899–7902
97. Wu X, Zhao Y, Ge H (2014) J Am Chem Soc 136:1789–1792
98. Li K, Wu Q, Lan J, You J (2015) Nat Commun 6:8404
99. Liu D, Liu C, Li H, Lei A (2013) Angew Chem Int Ed 52:4453–4456
100. Gartia Y, Ramidi P, Jones DE, Pulla S, Ghosh A (2014) Catal Lett 144:507–515
101. Cao W, Liu X, Peng R, He P, Lin L, Feng X (2013) Chem Commun 49:3470–3472
102. Wu X, Zhao Y, Ge H (2015) J Am Chem Soc 137:4924–4927
103. Yan S-Y, Liu Y-J, Bin L, Liu Y-H, Zhang Z-Z, Shi B-F (2015) Chem Commun 51:7341–7344
104. Lin C, Yu W, Yao J, Wang B, Liu Z, Zhang Y (2015) Org Lett 17:1340–1343
105. Wang X, Qiu R, Yan C, Reddy VP, Zhu L, Xu X, Yin S-F (2015) Org Lett 17:1970–1973
106. Ye X, Petersen JL, Shi X (2015) Chem Commun 51:7863–7866
107. Wu X, Zhao Y, Ge H (2014) Chem Eur J 20:9530–9533
108. Nakao Y, Idei H, Kanyiva KS, Hiyama T (2009) J Am Chem Soc 131:5070–5071
109. Li M, Yang Y, Zhou D, Wan D, You J (2015) Org Lett 17:2546–2549
110. Maity S, Agasti S, Earsad AM, Hazra A, Maiti D (2015) Chem Eur J 21:11320–11324
111. Miyazaki Y, Yamada Y, Nakao Y, Hiyama T (2012) Chem Lett 41:298–300
112. Nakao Y, Morita E, Idei H, Hiyama T (2011) J Am Chem Soc 133:3264–3267
113. Ruan Z, Lackner S, Ackermann L (2016) ACS Catal 6:4690–4693
114. Zheng X-X, Du C, Zhao X-M, Zhu X, Suo J-F, Hao X-Q, Niu J-L, Song M-P (2016) J Org Chem 81:4002–4011
115. Zhan B-B, Liu Y-H, Hu F, Shi B-F (2016) Chem Commun 52:4934–4937
116. Aihara Y, Chatani N (2016) ACS Catal 6:4323–4329
117. Yan Q, Chen Z, Yu W, Yin H, Liu Z, Zhang Y (2015) Org Lett 17:2482–2485